BEYOND EVALUATION USE

An Empirical Study of
Involvement and Use/Influence in
Multi-Site Evaluations

Report 2:
A Case Study of the Impact of the Advanced Technological Education (ATE) Program Evaluation

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Executive Summary

In November 2004, the National Science Foundation (NSF) awarded Drs. Frances Lawrenz and Jean King of the University of Minnesota a grant ("Beyond Evaluation Use") to study the relationship between involvement and use in multi-site evaluations. The overall question guiding this research is: What is the relationship between involvement in evaluation and use of the evaluation processes and outcomes in multi-site evaluations? The grant activities were aimed at describing and measuring how four NSF-funded program-level evaluations involved projects in the multi-site evaluation and, in turn, how this level of involvement affected the ways in which individuals, the projects, and the STEM education and evaluation communities used the process or results. NSF was the primary "client" for the evaluation. Therefore, the individual projects and the fields of science, technology, engineering, and mathematics (STEM) education and evaluation were secondary audiences, not directly the intended users.

Evaluation projects related to the following four NSF programs were included in the study: Advanced Technological Education (ATE), Collaboratives for Excellence in Teacher Preparation (CETP), Local Systemic Change through Teacher Enhancement (LSC), and Mathematics and Science Partnerships – Research, Evaluation and Technical Assistance (MSP-RETA).

Data were collected through an online survey distributed to evaluation project leaders[i] and evaluators, interviews with project leaders and evaluators, an interview with one of the NSF program officers responsible for the program at the time, an on line survey of NSF education project PIs and questions of journal editors. Additional data were collected through review of archival documents, publications, and reports produced by the evaluation projects and through reflective descriptions provided by the evaluation projects principal investigators.

This case study examines the relationship between involvement and use in the program-level evaluation of one of these programs – the ATE program evaluation project. Findings suggest that the ATE program evaluation had a moderate level of influence on the ATE projects, which was only slightly enhanced by the involvement of the projects in the evaluation. This case study presented evidence of influence of the program evaluation on both the projects and the field, despite low levels of involvement. Other variables that affected use included project perceptions related to the extent to which the evaluation was representative of their own project and the opportunities the program provide for interaction and collaboration with other projects and evaluators.

The case study showed a weak relationship between project involvement in the ATE program evaluation and the influence of the evaluation's processes and products on the projects. The weak relationship between involvement and use was primarily due to limited opportunities for project involvement, not evidence of use, which was available for the projects and the field, albeit to a slightly lesser extent.

Introduction

In November 2004, the National Science Foundation (NSF) awarded Drs. Frances Lawrenz and Jean King of the University of Minnesota a grant ("Beyond Evaluation Use") to study the relationship between involvement and use in multi-site evaluations. The overall question guiding this research is: What is the relationship between involvement in evaluation and use of the evaluation processes and outcomes in multi-site evaluations? The grant activities were aimed at describing and measuring how four NSF-funded program-level evaluations involved projects in the multi-site evaluation and, in turn, how this level of involvement affected the ways in which individuals, the projects, and the STEM education and evaluation communities used the process or results. NSF was the primary "client" for the evaluation. Therefore, the individual projects and the fields of science, technology, engineering, and mathematics (STEM) education and evaluation were secondary audiences, not directly the intended users.

The following four NSF programs were included in the study: Advanced Technological Education (ATE), Collaboratives for Excellence in Teacher Preparation (CETP), Local Systemic Change through Teacher Enhancement (LSC), and Mathematics and Science Partnerships – Research, Evaluation and Technical Assistance (MSP-RETA).

This case study examines the relationship between involvement and use in the program-level evaluation of one of these programs – the ATE program. The case study has three sections. First, it describes the history of the ATE and its program evaluation and the context within which the evaluation was conducted. Second, it presents the team's findings related to the relationship between project-level involvement in the evaluation and use of the evaluation processes and outcomes. And finally, it presents the findings about the extent to which the ATE program evaluation influenced the fields of STEM education and evaluation.

Methods

Data were collected through an online survey distributed to ATE project leaders and evaluators, interviews with ATE project leaders and evaluators, an interview with an NSF program officer responsible for the ATE program at the time, an online survey of NSF Principal Investigators (PIs), and a short survey of journal editors. Additional data were collected through review of archival documents, publications, and reports produced by the ATE program and program evaluation and through reflective descriptions provided by the ATE program evaluation PI.

ATE Project Leader and Evaluator Survey

The research team acquired names and email addresses for the PIs, co-PIs and evaluators of ATE projects. The team compiled lists from project rosters and meeting rosters provided by the program evaluation personnel. Although the lists contained the most recent email addresses on record, some names were missing email addresses. In such instances, a member of the Beyond Evaluation Use Grant (EUG) team searched the Internet for an email address. In all, the research team acquired names and email addresses for 493 ATE representatives.

All individuals with an email address received an initial invitation email with a description of the survey, an explanation of its use, and the Human Subjects approval number and phone numbers to call with questions. The email also contained a link to the online survey with a login and password. Two and four weeks following the first email, the team sent reminders to nonrespondents. The data were collected from August 10, 2006 through January 17, 2007.

Complete details of the survey's development, administration, and validation are presented in "Beyond Evaluation Use Report 1: Findings from the NSF Follow-up Survey." Survey findings presented in this report are taken from 57 close-ended items and four open-ended questions related to the involvement in and use of the ATE program evaluation.

Eighty-four of the initial 493 email addresses were returned as undeliverable and no valid email was found. Of the remaining 409 PIs and evaluators, 188 responded, resulting in a 46% response rate. Two respondents provided unusable data so their surveys were not included in the analysis. The majority of respondents (92.5%) were either PIs or project personnel.

Responses to open-ended items were reviewed by team members. When appropriate, responses may be used to support or exemplify findings from the close-ended survey items or the project interview data, which is described in the next section of this report. The following table shows the number and percentage of respondents who provided comments to each open-ended question or statement.

Table 1. Responses to open-ended items

Type of Comment	Comment[ii]	No Comment	Percent Respondents with Comments
Example of a negative influence of the program evaluation	21	180	11.7%
Example of factors that enabled use	82	104	44.1%
Example of factors that limited use	64	122	34.4%
Example of inappropriate use	6	180	3.2%
General comments about the program evaluation	39	147	21.0%

Nonrespondent Study. To check for nonresponse bias, the team conducted a nonrespondent follow-up survey after the online survey of the PIs and evaluators. We developed a three-question nonrespondent survey to address the topics of involvement and use, as well as to gather some information about what kept individuals from responding to the initial survey and reminder emails. The nonrespondent survey items were:

1. How involved were you in the ATE Program Evaluation?
2. How much impact did the ATE Program Evaluation have on you?

3. What was the main thing that kept you from responding to our initial survey request?

A random sample of nonrespondents to the survey was drawn using SPSS. The team sent each person an individually addressed email with a request for input on the three items. Two reminder emails were sent to each of the respondents who did not respond to the initial request. Then, if email contact did not solicit a response, a team member contacted the individual by telephone. Forty-nine percent of the nonrespondents (36/74) completed the nonrespondent survey.

The results of the analysis comparing the ATE respondents and nonrespondents showed no significant difference in the overall level of involvement reported (p=.91). However, the analysis of the impact question showed a significant difference between respondents and nonrespondents (p=.03). Means of the survey's involvement and use items were used to measure level of involvement and impact for respondents for comparison to the nonrespondents. The process of distilling the larger survey's multiple items related to use into a single question resulted in the choice of the word "impact" as a global concept to encompass questions about use of the process, instruments, and results of the evaluation. Therefore, the difference between respondents and nonrespondents may indicate the possibility of upward bias in the estimates of overall use for ATE survey respondents or it may also be an artifact of the difference between use, influence and overall impact. See Table 2.

Table 2. Results of nonrespondent study

Involvement Mean			**Use/Impact Mean**		
Respondents	Non-Respondents	p-value	Respondents	Non-Respondents	p-value
2.39	2.36	.91	2.50	2.06	**.03**

Interviews

Research team members conducted in-depth interviews with a purposefully selected sample of four project leaders and evaluators who had provided data on the online survey (the Project Leader and Evaluator survey) and had said they were willing to be interviewed. Survey respondents' responses to the involvement and use questions were used to identify four key types of projects: 1) high involvement, high use; 2) high involvement, low use; 3) low involvement, high use; and 4) low involvement, low use. These categories were determined based on relative reported levels of involvement and use, and interview participants were selected based not only on their responses to the survey but also to balance project leaders and evaluators and ensure diversity in the number of projects sampled.

The research team contacted interviewees contacted by email and invited to participate. Each interview was conducted with a single respondent and at least one EUG team member. The interviews followed a standardized, semi-structured protocol with flexibility in probing for

additional details and description as necessary. The length of the interviews ranged from 30 to 45 minutes. All interviews were taped and transcribed.

Based on a recommendation from the EUG advisory committee in fall 2007, the research team interviewed project leaders and evaluators who indicated high levels of use on the Project Leader and Evaluator survey. The goal of these interviews was to gain an in-depth understanding of high users, especially related to their level of involvement or any other characteristics that emerged. Survey respondents with the following characteristics were selected for the "high user" sample:

- Indicated a willingness to be interviewed,
- Averaged at minimum 3.0/4.0 on the 32 survey items that measured use and influence,
- Had neither been contacted for a previous interview nor been interviewed prior to the high use interview.

The team did not consider the involvement level or the roles (PI or evaluator) when selecting interviewees to be contacted. Rather, the 26 ATE survey respondents with mean use/influence scores from 3.0 – 4.0 were prioritized from highest mean to lowest and then contacted in stages about their willingness to participate. The five respondents who agreed to participate in the high user interviews were all PIs, had use means that ranged from 3.4 – 4.0, and involvement means of at least 3.0. Table 3 provides a description of the eight individuals interviewed in this study in terms of their levels of use and involvement and their status as evaluators, primary investigators, or other staff members.

Table 3. Description of interviewees

	Low Use	High Use
Low Involvement	Interviewee 3 (PI)	Interviewee 1 (PI)
High Involvement	Interviewee 4 (PI)	Interviewee 2 (Eval.) Interviewee 5 (PI) Interviewee 6 (PI) Interviewee 7 (PI) Interviewee 8 ((PI) Interviewee 9 (Eval.)

Frances Lawrenz, PI for the University of Minnesota project, conducted an additional interview with one of several NSF program officers who shared responsibility for the ATE program. This interview lasted approximately 25 minutes and was taped and transcribed.

Citation Analysis

The citation analysis used bibliographic methods to gather data about the use of program evaluation products by individuals within the STEM education and program evaluation communities. Citation analysis consists of tracking the number of citations other authors make to a published work. Web of Science, Google Scholar, and Google were used to collect citation

information from lists of evaluation products produced by the ATE program evaluation (see Appendix A and B for lists of the evaluation products and citing works). The list of the ATE evaluation products was initially compiled by reviewing the Western Michigan University ATE evaluation website. Products were defined as any publication, presentation, instrument, or other related material that were produced as part of the NSF-funded evaluation project. The PI of the evaluation reviewed the list for completeness and accuracy and suggested needed changes to the list.

The Web of Science, Google Scholar, and Google search engines were used to collect citation information from the evaluation product list. Searching was completed in late 2007 during the following periods: Web of Science August 27-30, 2007, Google Scholar September 4-6, 2007, and Google September 6 and 10, 2007.

Web of Science Searching	Google Scholar Searching	Google Searching
August 27-30, 2007	September 4-6, 2007	September 6 and 10, 2007

Data were collected manually by entering full reference information about the citations into an Excel spreadsheet that was later checked for accuracy and compliance with APA style and converted into an SPSS data file. Complete details of the rationale, development, administration, and validation of the citation analysis are presented in "Beyond Evaluation Use: A Citation Analysis of the Influence of the ATE, CETP, LSC, and MSP-RETA Evaluations on the STEM Education and Evaluation Fields, Report 6."

NSF PI Survey

As a measure influence on the field, the research team administered an online survey to a sample of 350 individuals randomly selected from the database of over 2000 currently-funded NSF Education and Human Development Directorate PI's. The survey elicited responses about PIs' perceptions of evaluations, people, and publications that might have been influential on the fields of STEM evaluation and education. The overall response rate was 54.7%. Complete details of the survey's development, administration, and analysis are presented in "Beyond Evaluation Use Report 7: Findings from the Survey of Influence of the ATE, CETP, LSC, and MSP-RETA Evaluations on the STEM Education Field."

Journal Editor Inquiry

A very brief email/telephone survey of editors and coeditors of six journals in the evaluation and STEM education fields was also conducted to determine if the editors were aware of any articles that had been published about influential STEM evaluations. Responses were obtained from three of six editors/coeditors. Complete details of the survey's development, administration, and analysis are presented in "Beyond Evaluation Use Report 7: Findings from the Survey of Influence of the ATE, CETP, LSC, and MSP-RETA Evaluations on the STEM Education Field."

Archival Review

The collection of archival documents for ATE was conducted primarily using electronic databases and search engines. The NSF website provided information on the ATE program as a whole, as well as on individual ATE projects. The archival documents included reports, grant products, survey instruments, journal articles, NSF grant solicitations, project summaries, and newsletters.

The principal source of information was the extensive collection available on the website of the Evaluation Center at Western Michigan University, including reports, fact sheets, survey items, grant products, brochures, handbooks, papers, and other products, and copies of, or citations for, several articles in scholarly journals. The team also searched the Internet using Google, several electronic indexes including Academic Search, WilsonWeb (the H.W. Wilson Information Retrieval System for the World Wide Web), ERIC (the Education Resources Information Center), and the NSF-sponsored OERL (the Online Evaluation Resource Library). Complete details of the document collection and analysis are presented in "Beyond Evaluation Use Report 6: A Citation Analysis of the Influence of the ATE, CETP, LSC, and MSP-RETA Evaluations on the STEM Education and Evaluation Fields.

ATE Program and Evaluation Background and Context

This section of the case study presents details of both the program and the program evaluation to provide context for the discussion of evaluation involvement and use.

ATE Program

Working under a Congressional mandate, the National Science Foundation (NSF) created the Advanced Technological Education (ATE) Program in 1992 to improve advanced technological education. NSF aimed to increase the number and quality of skilled technicians in the workforce[iii] thus improving U.S. competitiveness amidst growing global competition.[iv] Viewed as a conduit for preparing technicians, community colleges received most of the ATE funds to strengthen scientific/technical education and training capabilities in biotechnology, environmental technology, GIS, manufacturing, and telecommunications.[v]

ATE Program Facts

Period: 1993-present

Purpose: (1) Build capacity to provide advance tech education; (2) Increase number of skilled technicians.

Primary Target: Students & teachers in 2-year colleges and collaborating institutions.

Between 1994 and 2005, ATE funded 674 projects and centers, making grants to 345 unique institutions of which more than 200 were two-year colleges. Total cost of the program was approximately $350 million. ATE funded large Centers of Excellence (up to 5 million for 4 years) and smaller-scale projects (up to $300,000 for 3 years). ATE projects focused on improving technical education materials, enhancing technical instruction, and providing professional

development to faculty and teachers. Grantees established partnerships with high schools, two- and four-year colleges, businesses, government agencies, and professional societies to improve the nation's advanced technology workforce. [vi,vii] ATE centers could not be purely local; they required either a regional or national focus.[viii]

ATE Program Evaluation Background

In 1999, NSF funded the Evaluation Center at Western Michigan University (WMU) to conduct an external evaluation of the entire ATE program to collect data for ATE program reporting and to offer program improvement feedback to individual ATE projects.[ix] Arlen Gullickson, Principal Investigator, received two consecutive NSF grants: the initial grant of $1.3 million covered 1999-2002, and a subsequent $1.8 million grant covered 2002-2005.[x] Although some components of the ATE program evaluation are still ongoing, the scope of this case study in limited to the years 1999 – 2005. Consequently, activities that have occurred since 2005 are neither considered nor commented upon, despite their potential impact on involvement and use or influence.

To meet the reporting requirements of the Government Performance Results Act (GPRA), the ATE evaluation design focused on being "responsive to and meeting the needs of NSF." However, part way through the multi-year ATE program evaluation, NSF shifted its focus to highlight the need for evaluation products to be useful to STEM educators and evaluators.[xi]

ATE Evaluation Facts

NSF Grant Funding
•$1.3 million (1999-2002)
•$1.8 million (2003-2005)

NSF Goals for ATE Program Evaluation
•Provide evidence of quality of ATE program
•Inform program improvement

Evaluation Questions
•To what degree is the program achieving its goals?
•Is it making an impact and reaching the individuals and groups intended?
•How effective is it when it reaches its constituents?
•Are there ways to significantly improve the program?

Personnel/Expertise
•NSF 3-person mgmt team
•Experienced evaluation team
•Eval Advisory Committee

ATE Evaluation Questions

The evaluation assessed the impact and effectiveness of the ATE program by addressing four basic evaluation questions:

1. To what degree is the program achieving its goals?
2. Is it making an impact and reaching the individuals and groups intended?
3. How effective is it when it reaches its constituents?
4. Are there ways the program can be significantly improved?[xii]

ATE Evaluation Design

The evaluation focused on program-wide productivity and accountability with the primary audience being NSF and then Congress. It is important to understand that the evaluators structured the evaluation in a manner that would allow them to provide feedback to NSF. Consequently, the evaluation was not designed to necessarily influence the projects and centers.

Principal evaluation activities included: collecting descriptive data, developing an annual web-based evaluation survey to collect data over time, conducting site visits, drafting targeted reports on various aspects of the program, and providing formative evaluation information to NSF and to the projects and centers.[xiii]

The ATE evaluation featured annual web-based surveys and multiple site visits for gathering primarily qualitative information. The annual web-based surveys collected quantitative data on the activities, accomplishments, and effectiveness of ATE projects and centers for general accountability purposes; the site visits yielded data on the underlying drivers of program success including collaboration with partners, professional development, and project sustainability. The evaluation also included targeted studies on various aspects of the program and formative evaluation information to NSF and to the projects and centers.

Overall, the process of creating the survey framework and the question wording involved NSF program officers and an evaluation advisory group, but individual ATE grantees had little participation or input. The evaluation advisory group, comprised of individuals who worked directly with ATE projects and centers, helped conceptualize the survey and contributed to item development.

ATE Evaluation Components

The following is a brief description of each of the components of the ATE evaluation:

Annual web-based surveys. Web-based annual project surveys formed the backbone of the ATE evaluation. They provided common information across all sites about the program operations, the numbers and types of participants, and project activities and interactions.[xiv] At the time, no commercial web-based survey programs had been developed, so the WMU team decided to design and build its own online survey instrument, an expensive and time-consuming undertaking. Throughout this report, this survey will be referred to as the "annual web-based survey," not to be confused with this research team's online survey of the project leaders and evaluators, which will be referred to as the "online survey" or simply "survey" (i.e., "survey respondents reported…").

Site visits. Teams of experts visited 13 ATE sites (3 centers, 10 projects) and used prepared protocols to interview PIs, industry collaborators, faculty and students. They also reviewed materials, toured facilities, and observed classes to validate survey findings and gather

additional information. The site visit team wrote a report to each site, but NSF did not receive copies of these reports.[xv]

Site visit guide. Following the visits to 13 sites, the ATE evaluation team created a detailed manual or handbook to aid ATE projects and centers in how to conduct site visits. The document, "A Guide for Planning and Implementing Site Visits" written by Frances Lawrenz, Nanette Keiser, and Beth Lavoie, served as a hands-on guide for ATE projects in conducting effective site visits. [xvi]

Issue papers. The original ATE evaluation plan called for combining the site visit reports into a single case study. The evaluation team, the project advisory committee, and the NSF program officers agreed to replace the single case study with a series of issue papers to synthesize the survey data, site visit data, and existing research. Replacing the case study with issue papers illustrated the shift toward emphasizing evaluation products that would be more useful to STEM educators and evaluators. The evaluation team selected expert authors to write research-oriented issue papers designed to appeal to a diverse audience on the following topics: collaboration, dissemination, materials development, professional development, program improvement, recruitment and retention, sustainability, advisory committees, and evaluation. The papers went through a stringent review and revision process to ensure validity and interpretability. "The ATE Program Issues for Consideration" by Frances Lawrenz and Nanette Keiser summarizes the issue papers.[xvii] In addition, Wayne Welch authored three issue reports more directly related to evaluation. These reports were more directly used by NSF and less so by STEM educators and evaluators.

Targeted studies. The second ATE evaluation grant included funding for in-depth targeted studies on various aspects of the program. Reports on the four studies posted at the WMU Evaluation Center website are:

1. ATE Sustainability Study
2. ATE Business and Industry Study
3. ATE Materials Development
4. ATE Professional Development Reports[xviii]

Dissemination. The prolific ATE evaluation team produced more than 50 reports and publications which are archived at the WMU Evaluation Center website at http://www.wmich.edu/evalct. The website has reports from each of the annual surveys, site visit reports, issue papers, and other publications designed to be user-friendly and appealing to wide range of STEM educators and evaluators. Finally, the website offers a password protected online, interactive data viewer for use by ATE projects.

National Visiting Committee Handbook. All ATE grants who received a specified level of funding, which varied by year, required a National Visiting Committee (NVC) to provide advice, assess progress, and provide assistance.[xix] In 2001, NSF made a supplemental grant to WMU for a study to help improve the effectiveness of ATE advisory committees. The evaluation team had

trouble finding any ATE projects that would welcome a site visit to review their advisory committee practices. However, NSF stepped in and successfully encouraged projects with functioning advisory committees to welcome a site visit by WMU evaluators. The study resulted in the production of the *National Visiting Committee Handbook* to help those responsible for organizing an NVC, planning and conducting meetings, and for post-meeting activities.

Brochures. To meet NSF's goal of wide dissemination of the ATE evaluation data, the evaluation team produced tri-fold brochures to better reach ATE collaborators, administration, and other supporters. The brochures summarized the findings and recommendations from three targeted studies. A fourth brochure provided an advisory committee checklist for use by ATE projects and centers. The brochures were on the following topics: Advisory Committee Checklist, Improving Sustainability, Materials Development, and Recruitment and Retention.

Meta-Evaluation. Four external meta-evaluators observed and analyzed the WMU evaluation activities. Hartmann and Loizides (2001) evaluated the web-based ATE evaluation survey system. Migotsky and Stake (2001) evaluated site visits and issue papers. Pucel (2003) evaluated the 2002 and 2003 survey reports. The final meta-evaluation by Mark (2006) considered the entire evaluation. Overall these reports were positive and offered useful suggestions for improving evaluation activities going forward.

Evaluation in Retrospect[xx]

The WMU evaluation team worked hard to solicit input from both the projects and centers, as well as from the NSF program officers. Gullickson and the WMU evaluation team convened the ATE Advisory Committee and found it to be an asset to the ATE evaluation project. They also formed a special evaluation advisory group to provide expertise and insight in shaping the design of the ATE evaluation. These advisory groups provided assistance in constructing the evaluation processes and in conducting site visits.

The senior staff at NSF and both advisory groups provided substantial input and guidance on the evaluation design. This process included frequent communications with NSF staff. Gullickson noted that members of the advisory group, their reputations, and their extensive professional networks influenced NSF staff members, as well as the STEM and evaluation communities.

Over time, NSF's early hands-on involvement (e.g., attending meetings, contributing to decisions on topics ranging from mapping the survey framework to creating individual survey items) leveled off. Through fewer face-to-face interactions and reduced oversight of the evaluation, the NSF role evolved from active co-creation to a more review-and-comment orientation. In fact, an ATE evaluation official commented on the reduced impact of WMU evaluation reports because fewer NSF personnel read them. This arguably resulted in less influence among the rank and file project personnel to read the evaluation reports, which could result in reduced use of evaluation information.

In terms of lessons learned, the WMU evaluation team reported gaining an appreciation of clear work plans and enforceable contracts with project partners. They found the user-friendly tri-fold brochures easy to disseminate, and they reported that ATE project staff members preferred brief explanations and steps to take that will help them succeed.

The WMU evaluation team also reported that greater clarity in NSF guidelines resulted from their evaluation findings, specifically clarification of expectations related to sustainability. Even after the evaluation project itself had ended, NSF officials insisted that the annual web-based survey continue, underscoring the ongoing effectiveness and importance of the web-based survey. WMU staff interpreted this fact as an example of influence, in that the NSF officials found this information valuable and wanted to continue to receive these reports.

NSF Program Officer's Perspective on the ATE Program Evaluation

This section of the case study describes the ATE program evaluation from one NSF program officer's perspective, based on an interview conducted by Frances Lawrenz. The purpose of the interview was to better understand the agency's role in and the use of the ATE program evaluation from a program officer's perspective. We acknowledge that this data is based on the perspective of only one of several program officers. It is possible that other program officers may have stressed different points or ideas.

NSF Involvement. NSF felt they were collaboratively involved as guides but not really in control of the evaluation. Early on, program officers worked closely with Western Michigan on the annual survey, making suggestions regarding the number, wording, and types of questions. Survey feedback occurred after data were collected and results indicated possible misinterpretation of some of the questions by the projects. The following two examples demonstrate the ways in which NSF helped structure evaluation by encouraging Western Michigan to elaborate on their requests or expand their methods.

> In the later ones [annual web-based surveys] people were reporting godly numbers of collaborations. And we said, "Alright, somebody doesn't understand what collaboration is." So I think we pushed to get that defined.
>
> I think we also were involved fairly early on in the fact that we got the [annual web-based survey] reports and the reports didn't represent the program to us. And we thought there was a lot going on in the program which was not being picked up in the numbers. And so some of it then [was looked at it more in depth] the Western Michigan group looked at various parts of materials development [or] professional development- there were seven or nine reports on all five [major elements of ATE] which was an attempt to do some of it.

NSF also influenced the data interpretation, or more specifically, the presentation of the interpretation. In one case, the evaluators had reported that one third of the PIs did not do an activity, but the NSF encouraged them to focus on the positive – that two thirds did do it. Although NSF was involved in the development of the materials, they were careful to give the

evaluators at WMU control. For example, NSF attended advisory team meetings and annual PI conferences, but their role remained focused on guidance rather than instruction.

> Arlen had an advisory committee and Liz and I wanted to attend it. There was some issues with some of the advisory committee, but I think after we did it that everyone felt more comfortable. So it really was participatory in all parts and I don't think we stepped over the line. But, if we did, people pushed us back and said "you can't do that." But we tried to ask the questions and make sure questions got asked but we did not- we had no influence on the answers.

Influence on NSF. Although Western Michigan perceived NSF's interest waned over time, the NSF program officer felt that his interaction with the evaluators was "probably reasonably strong to begin with and has probably gotten stronger." For example, NSF read evaluation reports carefully throughout the evaluation. The reports helped NSF improve its response to the ATE projects as well as inform decisions related to improving the program. The program officer provided this example of improving his responses to the projects:

> …one of the things that came out [of the annual web-based survey] was that something like 25% of the participants said they received, I think it was either no communication or no communications on annual reports. And so …… I responded, tried to write a response to every annual report -sometimes short, but sometimes long.

In addition, the program officer stated that he learned about evaluation from his involvement in the program evaluation. He believes his response to evaluation is different now and he is more likely "to push the envelope" in terms of what he hopes will be achieved by the evaluation. In addition, the program officer allocates more time to providing individual responses to evaluation reports.

NSF Perceptions of Project Involvement. NSF thought the projects were only minimally involved overall throughout the life of the evaluation, but that the little feedback NSF received from the projects was, indeed, used to change the evaluation. Two examples illustrate this point:

> At each annual meeting he [the program evaluator from Western Michigan] has a table or section in which he invites people to come in and do it. We have also in our "getting started" workshop- conference workshop- essentially told people that you've got a year off, certainly new people don't answer [the survey] the first year. And that's I think another interaction we had which was to say the data were skewed by this [projects just beginning] and so we gave them [projects] one year. But we tell them to look at it [the survey] so they [projects] know what data to collect.

> I think certainly in the survey instrument to reduce the number of questions and to try to get some idea of what the questions really meant to them. I think they all [the projects] had an influence in that.

NSF Perceptions about the Influence on Projects. The NSF program officer felt there was some effect, mostly in terms of learning about evaluation and perhaps being pushed to more numerical types of measurements. The use of program evaluation data in project reports was one direct way to see how projects used the evaluation, but there was little other direct feedback saying how the program evaluation changed or impacted them.

Project-Level Involvement and Use

This section of the case study examines the relationship between project involvement in the program evaluation and use of the evaluation process and outcomes. It presents the online survey (the Project Leader and Evaluator Survey) and interview findings organized according to the following topics:

- Motivation for Involvement
- Levels of Involvement
- Perceived Effect of Involvement
- Use by Project Leaders and Evaluators
- Use by Projects
- Factors Affecting Use

For each topic, quantitative results from the online survey will be discussed first. Data from project interviews will be addressed next. Relevant examples from the open-ended online survey items may also be included.

Motivations for Involvement

Project leaders and evaluators were most likely to report that their engagement in all of the stages of the program evaluation (planning, implementation, and communication) had been voluntary. However, a sizeable group (32% - 38%) also reported that they felt involvement was required in all stages. Survey respondents were most likely to report feeling required to be involved in the final stage of the evaluation: communication of the findings (see Table 4). This finding is possibly a reflection of the type of data required by the annual web-based survey. The annual web-based survey required detailed reporting of the project's activities, especially in terms of the numbers of people served, activities, hours, etc. So, projects likely could have felt that communication of findings was required if they considered completing the annual web-based survey (which was solicited from all projects in the second year of funding) a way of communicating their own project's performance and activities.

Table 4. Motivation for Involvement in the ATE Program Evaluation

Survey Question: Which option best describes the motivation for your involvement during the:	I was required	I was asked	I volunteered	N
evaluation planning stage	31.9%	29.2%	38.9%	113
evaluation implementation stage	34.6%	28.5%	36.9%	130
communication of evaluation findings stage	37.7%	22.1%	40.2%	122

During the project interviews, PIs and evaluators acknowledged that their primary reason for involvement was that they felt that the annual web-based survey was required and, therefore, was the primary reason for involvement. One PI explained simply, "I felt it was expected as part of my grant award."

However, the expectation to complete the web-based annual survey sometimes created an interest in getting involved in other ways and for other, more personal, reasons. One project PI explained that their project's motivation to complete the survey changed over the years,

> Probably, initially, it was just because we had this project and it was one requirement that we should be meeting to do it. I would say in the latter years, it was more that, okay they sent us two emails they were looking for this and that and we should probably just go and do it. By that time we had done so many of them and knew what kind of data we needed to be getting that they weren't nearly as much driven because we felt an obligation more from we'll just do this because they're asking us to do it type situation. (ATE Interviewee 1)

Compensation for involvement in other activities also motivated project leaders and evaluators. Although WMU did not pay projects for answering the survey, PIs may have received money to either support the first set of site visits or attend special meetings about the program evaluation. For example, a project leader stated "…I'm always willing to talk and share our comments and they also did offer a reimbursement for participating." A survey respondent explained in an open-ended comment section, "When you asked about motivation, I responded that I was asked. In truth, I was asked, but I was also PAID."

Levels of Involvement

The online survey asked respondents to indicate their level of involvement during the planning, implementation and communication phases of the program evaluation. Involvement in the program evaluation was generally low and centered primarily on the annual web-based survey that was expected of all projects by their second year of funding, as supported by both the online survey and interview data. Survey respondents reported at least a little involvement in all stages of the evaluation; such as discussions that focused on planning the evaluation, data collection and interpretation, and communication of the results. Means in each of the 13

activities ranged from 2.2 – 2.7[xxi], however, each activity had at least one third of the respondents reporting that they were not at all involved (37.1% - 47.7%). The means and percentages for each activity vary little, a reflection of the evaluator-directed program evaluation. The focus of the ATE evaluation was on program-wide productivity and accountability with the primary audience being NSF and then Congress. Consequently, although WMU did create opportunities for some project involvement, the evaluation was not designed as a participatory evaluation centered on the concerns and needs of the projects.

Survey respondents indicated that the most extensive involvement occurred during evaluation planning, specifically in discussions that focused the evaluation (47.8% or mean=2.7). The most common way to be involved during the implementation was in data collection. Although the mean and standard deviation for data collection is equal to that of data interpretation (mean=2.4 and SD=1.3), data collection had the smallest percentage of respondents who reported no involvement. During the final stage of the evaluation, almost half of the respondents (47.5%) reported that they had nothing to do with writing evaluation reports, rather the most common activity during this stage was the review of these reports, with 65.2% (mean=2.6) saying they were involved at least a little in reviewing reports for accuracy and completeness. Tables 5 - 7 below show the distribution of responses for each activity.

Table 5. Involvement in evaluation planning activities

Survey Statement: I was involved in…	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**N**	**Mean**	**SD**
the discussions that focused the evaluation.	37.1%	5.1%	10.1%	47.8%	178	2.7	1.4
identifying evaluation planning team members.	47.7%	5.2%	9.8%	37.4%	174	2.4	1.4
developing the evaluation plan.	41.2%	9.0%	14.7%	35.0%	177	2.4	1.3

Table 6. Involvement in evaluation implementation activities

Survey Statement: I was involved in…	No	Yes, a little	Yes, some	Yes, extensively	N	Mean	SD
developing data collection instruments.	43.3%	11.2%	16.9%	28.7%	178	2.3	1.3
developing data collection processes.	41.0%	12.4%	19.1%	27.5%	178	2.3	1.3
collecting data.	36.1%	18.9%	18.9%	26.1%	180	2.4	1.2
reviewing collected data for accuracy and/or completeness.	42.1%	12.4%	14.6%	30.9%	178	2.3	1.3
analyzing data.	46.4%	11.2%	14.5%	27.9%	179	2.2	1.3
interpreting collected data.	42.7%	10.1%	15.2%	32.0%	178	2.4	1.3

Table 7. Involvement in evaluation communication activities

Survey Statement: I was involved in…	No	Yes, a little	Yes, some	Yes, extensively	N	Mean	SD
writing evaluation reports.	47.5%	12.2%	11.0%	29.3%	181	2.2	1.3
reviewing evaluation reports for accuracy and/or completeness.	34.8%	11.0%	16.6%	37.6%	181	2.6	1.3
presenting evaluation findings.	40.8%	10.1%	11.2%	38.0%	179	2.5	1.4
developing future project plans based on evaluation results.	41.6%	5.6%	16.3%	36.5%	178	2.5	1.3

The interview data suggest that project leaders and evaluators considered submitting the annual web-based survey as the primary way in which they were involved, as it was mentioned by all interviewees. This finding is not surprising given the relatively high levels of involvement in data collection reported by survey respondents. However, the interview data suggest that submission of the survey by itself did not engender feelings of involvement. One PI explained that she had no involvement in the program evaluation except for filling out the survey, which she did only because it she felt it was required. Consequently, the PI qualified her level of involvement as only "a little."

A project evaluator, who described her involvement with the explanation "I'm not sure, but I certainly can't say much," knew of the annual web-based survey questions and provided some information so that the project could complete it. However, this evaluator did not expect the

project staff to reflect on or use the data in any way. Although this same project evaluator completed a questionnaire about the program evaluation at an annual meeting as well, she still considered her involvement as minimal.

Perceptions of involvement appeared to strengthen for project leaders who also provided input on question development or were asked to present at a meeting. However, even the PIs who got involved in other ways, reported only "some" involvement. For example, a project PI said:

> In addition, this past year, I was contacted by Western Michigan to review the questionnaire (the survey) and provided feedback to them as to whether the questions were appropriate and if I felt they were providing the proper feedback for that…Again, we're probably involved some. I mean, our involvement entails filling out the annual survey…. Also, the feedback [provided to WMU on the survey], certainly. (ATE Interviewee 1)

Attendance at annual meetings was mentioned by every interviewee as way in which they were involved. However, the attendance and participation at the meetings was not always directly related to the program evaluation. For example, only one interviewee completed a questionnaire about the program evaluation (response to questions, timing, etc.), the other interviewees' experiences only indirectly related to the evaluation. One project PI explained,

> Yes, we've attended PI meetings annually. We've showcased presentations of what we've done; we participated in round table discussions; and we have participated in pre-pamphlets workshops. We find the annual PI meetings extremely beneficial and we look forward to going to them every year… I believe we sat in on a round table discussion maybe a couple of years ago. During the showcase presentations, we've met individuals from…I believe we've met Arlen [Gullickson], but we've met people from Western Michigan; so yeah, we've met those who put on the annual survey. (ATE Interviewee 1)

Another PI described a presentation he gave at an annual meeting, during which he mentioned the evaluation,

> I did a workshop on evaluation at one of their annual meetings….And we did address the Western Michigan stuff… I believe there was someone from Western Michigan, but I'm not positive on that. (ATE Interviewee 4)

Perceived Effect of Involvement

Survey results suggest that project leaders and evaluators perceived their input was valued (among those who felt they could comment about the use of their input), but interview data were less conclusive. For those project managers and evaluators who did provide input, it appears that they perceived their involvement as important, with at least 49.1% indicating their input was both considered and used extensively and over 95% reporting that it was used at least a little. Input regarding communication of the evaluation findings was the least likely to

be reported as being used, but the percentage who thought their input was not used was still extremely low at 4.6%. Therefore, it appears that the survey respondents felt that their input was valued.

Table 8. Consideration and use of project leader and evaluator input

Survey Statement: The input I provided during the…	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**N***
evaluation planning stage was taken into consideration.	1.7%	11.1%	31.6%	55.6%	117
evaluation implementation stage was taken into consideration.	3.4%	12.7%	31.4%	52.5%	118
communication of evaluation findings stage was taken into consideration.	2.6%	12.2%	32.2%	53.0%	115
evaluation planning stage was used.	1.8%	15.3%	32.4%	50.5%	111
evaluation implementation stage was used.	1.8%	15.5%	33.6%	49.1%	110
communication of evaluation findings stage was used.	4.6%	11.1%	32.4%	51.9%	108

*Total N is smaller because some individuals responded that they did not provide input or they did not know if their input was considered or used.

Interviewees seemed unable to discern the ways in which the WMU evaluation team used their input. One project manager explained that his input basically consisted of positive feedback and the fact that nothing changed reflected use and consideration of his input. “Again, our input was that it looked like that they had things pretty much under control, so I believe our input was used.” (ATE Interviewee 1) Other interviewees simply did not know and responded to the question with simple, “I have no idea.” (ATE Interviewee 3)

Although survey responses indicate that project leaders and evaluators perceived that their input was valued, it is important to recognize that this finding reflects the views of those respondents who felt both involved (at some level) or that they were able to comment on how the input was considered or used. In light of the smaller number of survey respondents who chose to answer this set of questions in Table 8, it is possible that the interviewees’ lack of comments regarding how their input was considered or used reflects the perception of the survey respondents who chose not to respond to this set of questions.

Perceived Use by Projects Leaders and Evaluators

The survey asked project leaders and evaluators to reflect on the extent to which the ATE program evaluation influenced their 1) knowledge and understanding, 2) skills, and 3) beliefs about evaluation. Additionally, project PIs were asked about their use of the program evaluation process and findings to make decisions about their project or in other projects. The

interview questions solicited examples of use or impact on the individual and the project. In general, results suggest that the evaluation did influence project leaders and evaluators; in some cases extensively. However, means for each area of influence never exceeded 2.7 on a 4.0 scale (1=none, 2=yes, a little, 3=yes, some, 4=extensively). The highest percentages of extensive influence were related to beliefs about the planning, implementation, and communication stages of an evaluation.

Influence on Knowledge and Understanding

The majority of the respondents reported that they perceived the program evaluation increased their knowledge or understanding of evaluation (all stages), STEM education and evaluation, and their own project at least a little. The largest percentages of respondents categorized their increase in knowledge as "some" (31.9% - 37.6%). Almost a quarter reported they learned an extensive amount about their projects. Respondents were least likely to report gaining knowledge about STEM education, with almost one third reporting no influence (mean = 2.3). Means for other areas ranged from 2.4 to 2.6, indicating low to moderate levels of influence. For complete results, see Table 9 below.

Table 9. Influence of the ATE program evaluation on project leaders' and evaluators' knowledge and understanding

Survey Statement: The evaluation increased my knowledge/understanding of…	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**N**	**Mean**	**SD**
how to plan an evaluation.	22.3%	25.0%	34.2%	18.5%	184	2.5	1.0
how to implement an evaluation.	21.2%	27.7%	34.8%	16.3%	184	2.5	1.0
how to communicate evaluation findings.	25.7%	24.6%	35.5%	14.2%	183	2.4	1.0
Science, Technology, Engineering, and Mathematics (STEM) education.	31.9%	22.0%	31.9%	14.3%	182	2.3	1.1
STEM education evaluation.	23.2%	22.7%	37.6%	16.6%	181	2.5	1.0
my project.	22.1%	18.2%	35.9%	23.8%	181	2.6	1.1

Influence on Skills

Although the program evaluation's influence on skills reflected a similar pattern to its influence on knowledge and understanding, overall the levels are lower, as shown by lower means and slightly higher percentages of respondents reporting no influence (please see Table 10). The pattern was similar in that extensive influence of both types (skills and knowledge and understanding) was most likely to be related to the respondents' projects, as indicated by the percentage reporting extensive influence and the highest mean of 2.5. Additionally, STEM education was the area with the least amount of influence, as indicated by the lowest mean and highest percentage reporting no influence.

Table 10. Influence of the ATE program evaluation on project leaders' and evaluators' skills

Survey Statement: The evaluation improved my skills…	No	Yes, a little	Yes, some	Yes, extensively	N	Mean	SD
in planning an evaluation.	20.9%	29.7%	33.0%	16.5%	182	2.5	1.0
in implementing an evaluation.	21.4%	31.9%	32.4%	14.3%	182	2.4	1.0
in communicating evaluation findings.	25.8%	27.5%	33.0%	13.7%	182	2.3	1.0
as a STEM educator.	38.3%	22.4%	29.5%	9.8%	183	2.1	1.0
as a STEM education evaluator.	26.7%	28.3%	34.4%	10.6%	180	2.3	1.0
for working on my project.	23.6%	23.1%	35.2%	18.1%	182	2.5	1.0

Influence on Beliefs

Results in Table 11 suggest that the program evaluation had a stronger influence on beliefs related to the importance of evaluation rather than beliefs related to the importance of STEM education, STEM education evaluation, and their projects. There appears to be a definite distinction between beliefs about the evaluation, beliefs related to STEM and beliefs about individual projects, although each area always had a majority of respondents reporting at least a little influence.

Table 11. Influence of the ATE program evaluation on project leaders' and evaluators' beliefs

Survey Question: The evaluation increased my belief in the importance of…	No	Yes, a little	Yes, some	Yes, extensively	N	Mean	SD
planning an evaluation.	17.4%	23.4%	31.0%	28.3%	184	2.7	1.1
implementing an evaluation.	16.9%	19.7%	36.1%	27.3%	183	2.7	1.0
communicating evaluation findings.	19.0%	23.4%	31.0%	26.6%	184	2.7	1.1
STEM education.	29.9%	19.0%	32.1%	19.0%	184	2.4	1.1
STEM education evaluation.	25.5%	22.3%	33.2%	19.0%	184	2.5	1.1
my project.	25.7%	15.3%	33.9%	25.1%	183	2.6	1.1

It is important to note that the influence may be a negative reflection on the program evaluation in that because it did not do something (e.g., give enough advance notification), the respondent learned that it was important to do it differently. To address this issue, survey respondents were asked whether the evaluation had a negative impact of their beliefs about evaluation and, if so, to provide an example. Only 11.7% (21/180) reported that the program had a negative influence. If respondents answered "yes, it had a negative influence" they were asked to explain their response. Negative influences generally related to the resources necessary to complete the evaluation and the perception that the evaluation was too numbers-driven. A few examples follow:

> I became more aware of the complexity. The process to try to develop an evaluation has been painful. I don't believe we adequately framed the problem. The goal of the evaluation is fuzzy. It is not NSF's fault. We Community Colleges don't have consensus on our mission, or even on qualitative and quantitative measures of evaluating our value-add, much less the success of innovative ATE programs.

I lost time from primary role as an instructor, and it created extra stress.

This became a process of "bean-counting" and did not reflect the greater richness that we are trying to seek through our center.

The WMU ATE evaluation was too data element oriented. As WMU discovered the breadth and variety of ATE activities the online survey only grew and grew in an attempt to capture every project activity in online forms parsed into data bases.

Use in Another Context

For those respondents who participated in another evaluation after the ATE program evaluations, over 80% used what they learned from planning and implementing the evaluation in another context. Take-aways from the planning of the evaluation appear to be the most influential, with the lowest percentage of respondents indicating no use and the highest percentage reporting extensive use. Even the least used aspect of the evaluation, the data collection instruments, were used at least a little extent by over 60% of the respondents who participated in another evaluation subsequent to the ATE program evaluation. Please see Table 12.

Table 12. Use by project PIs of knowledge, products, processes in other contexts

Survey Question:	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**N**	**Mean**	**SD**
I used what I learned from planning the ATE program evaluation in another evaluation.	17.6%	26.5%	36.8%	19.1%	68	2.6	1.0
I used the evaluation plan from the ATE program evaluation as a model in another evaluation.	32.4%	19.1%	32.4%	16.2%	68	2.3	1.1
I used what I learned from implementing the ATE program evaluation in another evaluation.	17.9%	34.3%	32.8%	14.9%	67	2.4	1.0
I used data collection instruments from the ATE program evaluation in another evaluation.	39.7%	20.6%	29.4%	10.3%	68	2.1	1.1
I used what I learned from communicating evaluation findings from the ATE program evaluation in another evaluation.	27.9%	30.9%	26.5%	14.7%	68	2.3	1.0

Use of Findings

The survey asked only the PIs to report the ways in which they used the program evaluation findings, with the understanding that project leaders had more authority than evaluators to do so. In general, although there is ample evidence of use of findings for decision making, with anywhere from 60% - 69% reporting that the program evaluation helped make decisions, changes, or plans regarding that project to at least a little extent (means ranged from 2.2 – 2.4).

Of all of the use and impact related questions, only two had a majority of respondents who indicated that no use occurred, and both of those related to use of findings for seeking funding and promoting additional evaluation (please see Table 13).

Table 13. Ways project PIs used ATE program evaluation findings

Survey Statement: I used the ATE program evaluation findings to…	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**N***	**Mean**	**SD**
make decisions about the future existence of my ATE project (e.g., cancel or continue).	39.7%	15.2%	30.5%	14.6%	151	2.2	1.1
develop future plans for my ATE project.	29.8%	19.9%	32.5%	17.9%	151	2.4	1.1
increase the attention given to my ATE project.	37.5%	17.1%	29.6%	15.8%	152	2.2	1.1
meet contractual and/or legal requirements of my ATE project.	35.8%	12.6%	26.5%	25.2%	151	2.4	1.2
enhance my organization's commitment to my ATE project.	40.7%	14.7%	30.7%	14.0%	150	2.2	1.1
seek funding for my project(s).	54.7%	12.0%	20.7%	12.7%	150	1.9	1.1
promote additional evaluation of my project(s).	55.0%	16.6%	19.9%	8.6%	151	1.8	1.0
make changes to my project(s).	30.9%	23.0%	34.2%	11.8%	152	2.3	1.0
help plan a new project(s).	36.8%	22.4%	26.3%	14.5%	152	2.2	1.1

*Only project PIs were asked this question, hence the lower N.

Comparisons Among the Types of Influence

As shown in Table 14 data from the survey suggest that the ATE program evaluation had the greatest influence on project leaders and evaluators' knowledge and understandings, followed closely by influences on their skills, beliefs, and use in other contexts. To a lesser extent, Project PIs used the program evaluation in a variety of ways, including planning future projects or seeking funding for the project (this is discussed further in the following section labeled Use by Projects). It must be reiterated that response bias may exist for these evaluation "use" questions. The nonrespondent survey data suggest that ATE respondents may have higher levels of use and influence of the program evaluation than individuals who did not respond to the survey. Therefore, it is possible that the levels of use reported below are somewhat higher than the use levels in the entire ATE project population.

Table 14. Influence of program evaluation on project leaders and evaluators

Type of Influence	N	Mean	SD
Influence on knowledge/understanding (q25-30)	184	2.5	.9
Influence on skills (q31-36)	183	2.3	.9
Influence on beliefs (q38-43)	184	2.6	1.0
Use of knowledge, products, processes in other contexts (q45-49)*	68	2.3	.9
Ways of using findings (q50-58)**	152	2.2	.9

*items asked only of those reporting having conducted a subsequent evaluation.
**items asked only of project PIs.

The interviewees reported some to extensive impact of the program evaluation on themselves personally, although the majority of the comments were about the impact of the evaluation on the projects and field. The most notable finding related to the personal impact of the evaluation concerned influence on an evaluator's approach to research.

> Yes; so now, I can tell you that it absolutely impacted my own research, because it pursued me and pushed me to go farther, to understand how to capture all the things that I think these sites are doing. I think it pushed my own research, it pushed me to look at research literature a great deal more-I wouldn't have otherwise-and so I can honestly tell you their lack of breadth was useless for me, because I do it in my own research. (ATE Interviewee 2)

Despite the negative undertones of this following comment, the lesson learned became a positive outcome.

> The lasting effect--and I'm going to say on me, because I don't think it has a lasting effect on the ATE program--is that it forced me to look robust measures of student achievement and implementation success. Its lack of robustness pushed me. As far as I'm concerned, that's been quite nice. It's made me a better researcher. (ATE Interviewee 2)

One project leader credited the program evaluation with teaching him about evaluation, but also acknowledged that some of the lessons were not a good reflection on the evaluation.

> I would say I definitely learned some new evaluation skills…Some of the things I liked was the fact that is was web-based survey. They gave you the ability to start and stop the survey in the middle [and] you could delegate the survey to other people. Those are some of the positives. I thought they tried to use evaluative techniques too extensively from the standpoint of asking the same question in multiple different ways to see if you'd answer it in the same ways. Well, having been doing evaluation for most of my

> career, those became very obvious and, what the anticipated reasons behind that was, which I found a little bit insulting as a professional educator…. (ATE Interviewee 4)

Use by Projects

Online survey items focused on the ways in which project leaders and evaluators were influenced by the program evaluation, including the extent to which they used the process and findings to make decisions about projects. The interview data allowed for deeper understanding about the extent to which the program evaluation influenced the projects. Interviewees perceived the impact of ATE program evaluation on projects was lower than the personal impact described in the previous section. Overall, two themes emerged related to ways in which the program evaluation influenced the projects. First, at least two projects used the survey to help guide project activities, especially related to project evaluation. Two project leaders provided the following examples:

> The data we provided to Western Michigan, really a lot of times came from our external evaluator. We basically used that and also to assist the external evaluator. Okay, here's some questions we need to obviously find answers for, so that helped guide our external evaluators search for the data and things like that. But you know it was sort of a twofold process that we used what our external evaluator provided to us and obviously our own knowledge of our own project, but then used it to guide some of the questions we might ask in subsequent years for our external evaluator… Definitely sparked conversations with our external evaluator, within our own team, and between different projects that, you know teams of different projects. Our grant area tried to work as a collaborative. (ATE Interviewee 4)

> Well, again, I think it helps us as a grant person and a PI and co-PI, to insure that…well, it's a self-evaluation tool for us, to make sure that we are doing what we said we were going to do, what are the pluses and minuses; it keeps us on track and helps us simulate that data and we can use it as a benchmark. (ATE Interviewee 1)

The second impact related to the time it took to complete the survey. For example, one evaluator explained that the only impact the evaluation had on the project was time:

> It did impact them [the project] so I'm going to say one or two [none or a little], if impact means did it have anything to do with their activities, like they had to do a lot of work to get the stuff to Western Michigan in the form that Western Michigan wanted it. In that respect, it had an impact because it consumed time. As far as changing activities at the site, or being the basis for reflection, thinking of an internal assessment at the site, I'd say that had no impact. (ATE Interviewee 2)

Although the final research question describes the factors that limited influence of the evaluation, two interviewees provided reasons as to why the impact specifically on the projects was so low. Two interviewees expressed a lack of interest in the data because it was seen more

as a requirement than a culmination of helpful data. For example, a project leader who reported no impact on the project explained:

> I did it alone, sitting in front of the computer, using the data I had collected during the course of the project…We had something we were doing, and we needed to do, and the evaluation had no impact. (ATE Interviewee 3)

A project evaluator described how the requirement pressured the project in a negative way:

> I did not hear once of any of the projects using that information. I didn't hear of the projects saying that this was important information, they needed to get it right when they reported it to Michigan; there was some inflation -- admitted inflation -- in what they sent to Michigan because the theory of the project was that we were to make it look good, and no one ever discussed how that information came back to them, or how it was useful to them…. There was a lot of conversation about the amount it costs and the little value it produces, and so several of the sites were wondering why they just couldn't have some of that money, rather than pay people to make them write more reports. (ATE Interviewee 2)

Factors Affecting Use

Survey respondents and interviewees identified or suggested a number of factors that both enabled and limited the impact of the program evaluation. Collaboration with colleagues and the evaluation team at Western Michigan was the most common enabling factor noted by survey respondents, and was also mentioned in one of the interviews. Survey respondents also noted that the increase in awareness about project activities and impact, as well as NSF expectations, helped initiate use. Factors that limited use included lack of resources (time, money, and experience), poor communication, and the perceived misalignment between the annual web-based survey goals and individual project goals.

Enabling Factors

Collaboration. In a broad sense, collaboration facilitated use among ATE project leadership and evaluators. Survey respondents noted specifically that working with the Western Michigan team and other outside evaluators provided valuable feedback to the project. The feedback was given in such a manner that allowed for questioning and growth, which in turn fostered use. For example, one respondent explained:

> The individuals from the Evaluation Center were highly motivated and skillful. They found areas where we could improve and suggested ways to go about it. They discovered some things that we were doing well without our knowing it (Open-ended Survey Response).

One project leader who was interviewed explained that he and other ATE grant recipients at his institution decided to collaborate on how to gather the program evaluation data so that all three projects could make use of one external evaluator. Other respondents cited the collaboration which grew from interactions with colleagues at annual meetings and training sessions as an important factor enabling use, "It was helpful to share ideas with colleagues who are also evaluators. This cross pollination was perhaps the best part of the process as we met and communicated at ATE [meetings]." Collaboration with outside businesses was also seen as beneficial, with one respondent noting that "The coming together of both academic and businesses was by far the most exciting experience to the evaluation process."

Increased Awareness. Survey respondents also credited the evaluation with increasing their awareness about the project in such a manner that it facilitated use. Armed with more knowledge about the extent to which the grant was meeting its goals, project leaders sometimes felt more justified in their decision-making. For example, a survey respondent wrote:

> The evaluation process enabled me to affirm that the objectives were achieved, related activities were carried out accordingly, and progress of all significant aspects of the project were successfully attained as planned with possibly an even greater positive impact than originally projected (Open-ended Survey Response).

A few other respondents noted that the program evaluation also increased their awareness about NSF expectations. One respondent explained, "Completing the forms annually drew my attention to collecting specific data on my project and reminded me what NSF might be looking for to show that the program is a success." The awareness instilled confidence about securing other NSF grants, "The evaluation report helped us in identifying and correcting weaknesses in our program. It was influential in helping us get an additional NSF grant."

Challenging Factors

Lack of Resources. Survey respondents indicated that the general lack of resources for the program evaluation activities limited the influence of the ATE program evaluation. Although one respondent listed all three together, "Funding, time, and the difficulty of collecting quantitative data," it was more common for survey respondents to list only one (e.g., "time constraints," lack of prior experience," or "money"). The example below illustrates how this challenge affected the project:

> The reality that it is difficult for many participants to have adequate time to devote to revision and reactions to the evaluative process. It is a challenge when you are person that is constantly calling participants to the table to meet on the many facets of the grant including the evaluation process (Open-ended Survey Response).

During one of the interviews, a project leader offered this explanation about the impact of the lack of resources:

> I felt like the survey was a little bit long and what happens when a survey gets a little bit long I think it suffers on the data that's provided. Because when the survey is going to 5,6, 7 hours to do, well, I think that impacts the quality of what you may get back from some individuals because they may not spend the 5-6 hours that it requires….I think the length of the survey was a real concern even though I realize the projects have been funded fair amounts of money, but to do the program effectively you're out there doing the project and not necessarily to do another survey when you're already doing your own survey and evaluation and things like that. (ATE Interviewee 4)

Misalignment of Evaluation Indicators and Project Goals. The second factor that emerged from the survey respondents and interviewees addressed an apparent weakness in the annual survey. Respondents and interviewees perceived that the survey measured project effectiveness in different ways than the projects wanted to measure it. One interviewee described her concerns about the survey:

> I was concerned that we would paint ourselves to be so uni-dimensional, rather than multi-dimensional….it concerned me that the site absolutely lived and died by those benchmarks, and the sites would get themselves into these real quandaries about not meeting those benchmarks when, in fact, I think the benchmarks were wrong. (ATE Interviewee 2)

Survey responses highlighted similar concerns and explained in more detail the possible negative consequences of this misalignment.

> The questions were so general, and equivocating, that we had a hard time getting a handle on how to measure things, and how to measure the same things again looking for impact. I'm not convinced that an evaluation tool that is designed as 'one size fits all' really fit well enough to be measuring meaningful things in terms of the specific projects. Obviously, from my answers above, the evaluation wasn't a waste of time, and I'm not sure how it could have been done better, but I have some misgivings about accuracy. This might be the common problem of 'hard science meets social science' and the ways of measuring things differ significantly (Open-ended Survey Response).

Although survey respondents felt "that enough of the evaluation questions provided me with information I could use in other ways," several respondents and one interviewee completed the survey because it was required. For example, one survey respondent explained:

> It was clear we had a one-shot project, write up materials, try to get them published and adopted. It was clear after awhile that publisher were not interested. So use was limited to the workshop groups, a small set. Evaluation became a necessity in reporting back to NSF on what happened, not a tool in future planning (Open-ended Survey Response).

One interviewee felt that her project may have been so different from the mainstream that the evaluation results simply didn't apply to her project. This misalignment in a way seemed to create a disinterest.

> I guess because we had a very clear goal in our grant; it went the way we thought it was going to go; we reached a happy conclusion; we used the money well, but as we had anticipated using it. I wasn't waiting to hear about someone else's opinion about how we were doing things. (ATE Interviewee 3)

High Users' Perspectives

The following is a presentation data from interviews with five project leaders who the research team designated a high user based on his or her mean use/influence score on the online survey. Interviewees were selected based on mean use/influence score only, without consideration of involvement levels or role. Consequently, the five high user interviewees were all highly involved (mean involvement score equal to or greater than 3.0/4.0) and all project leaders and no evaluators. The presentation of the high user interview data is organized around the same six topics above:

- Motivation for Involvement
- Levels of Involvement
- Perceived Effect of Involvement
- Use by Project Leaders and Evaluation
- Use by Projects
- Factors Affecting Use

Motivation for Involvement. All PIs were personally motivated and expressed a desire to be involved. Two primary reasons for their involvement emerged from the interviews: 1) to learn (about evaluation or the survey itself) or, 2) to share project details with others. One interviewee explained that she felt that the learning about other projects and sharing details of her own was part of her personal responsibility to the program itself.

> It's part of our responsibility. We're working within this grant and we need to be able to go out and share information with other people and learn things from other people so that we can make sure that our program is doing what it needs to do and get ideas to make it better and to be able to share what we've learned with other people. (ATE Interviewee 7)

Beyond the responsibility aspect, another PI offered that she felt the sharing provided some benefit to her project. She explained, "I'm pretty proud of the work we've done here and the progress we've made and so it was nice to be able to fill our a survey that got some other recognition." (ATE Interviewee 9)

Levels of Involvement. The interviewees reported mixed levels of involvement – despite the survey results indicating that they all had a mean involvement score of at least 3.0 out of a possible 4.0. However, the basic activities of each PI were not drastically different. In two cases, the activities were exactly the same, but each reported different levels of overall involvement – one low and one extensively. All of the interviewees reported that they had completed the annual web-based survey and all but one (albeit the lowest level of involvement) attended PI meetings. Those who reported higher levels of involvement seemed to interact more with the evaluation, it's people, and the data collection associated with survey. Other ways of involvement included volunteering to pilot test or serve on a committee, studying and reporting results to others. In response to the question of overall involvement, one highly involved PI interviewee described her answer this way:

> Involved extensively. I always felt like I was sounding board and therefore I've always been – I've always felt like I was very involved. That it was partly mine. I have enormous respect for it and am very fond of all the people involved in it so I feel like they're people that I learn from constantly. (ATE Interviewee 6)

Perceived Effect of Involvement Use by Project Leaders and Evaluators. Although the interview protocol included a question about the extent to which interviewees felt that their input was or wasn't used by the program evaluation team, only two of the five interviewees were actually asked this question. The interviewer skipped this question when it seemed apparent that the there was a lack of input beyond the annual web-based survey about which the interviewee could comment. Both PIs who were asked, responded that yes, they felt the input had been used. The most involved PI, who explained that she felt she had "grown up with that survey" and recently sat on an advisory committee related to the annual web-based survey, offered the following description of how her the Western Michigan team considered input:

> The team definitely listened to and responded to the things that I brought up and that others did too, I'm sure. Unless there were some real reasons why they couldn't. Like, for instance, if you just felt the survey was too long and asked too many question about too many different things. If they were responding to what NSF wanted, there was no way they were going to lop off a section. (ATE Interviewee 6)

Use by Project Leaders and Evaluators. The personal impact on PIs ranged from low to high. Those who reported higher use explained that the evaluation broadened their perspective by shifting their focus from project-level to program-level. This new perspective helped PIs to see how their local projects fit into the national scheme.

Two interviewees specifically mentioned that the ATE program evaluation strengthened their views on the importance of evaluation.

> Because I am working with so many desperate groups and that tended to keep me focused somewhat on what I was doing rather than looking beyond. So, I really felt that

> when I went to the survey and went to the evaluation groups that it gave me a broader perspective. (ATE Interviewee 9)

Only one of the two interviewees who reported only a little impact noted that the program evaluation did help him build new evaluation skills. However, the only personal effect the remaining interviewee could report was that she felt a sense of gratitude at having completed the task.

Use by Projects. Projects used the program evaluation in various ways with moderate levels of impact. Interviewees reported using the annual survey to guide the project evaluator in the development and implementation of the local evaluation. One project created data collection processes to simplify the annual task of the web-based survey – the development of which was significant enough to merit an overall rating of program evaluation's impact on the project as "some."

> …we had to institute the process to get that [survey] information and be prepared to do this because it took us way too much time to try and find out that information and gather it and straighten it out….It took us way too long to gather it before but if you look at the process now, we'll know just what to do in the future. (ATE Interviewee 5)

Several projects, including the only project whose PI reported "a little" impact, used the program evaluation results to communicate and garner support from different stakeholder groups. One interviewee explained:

> It's such a weird thing to describe what we do and it's hard to understand so I send them [the evaluation] data and information about our school so that they'll know what we are and what we do. (ATE Interviewee 7)

In addition, the results informed project decision-making by showing the projects where to focus theirefforts. In two cases, the projects used the results to improve the project by solving problems.

Factors Affecting Use. Interviewees agreed on a few primary factors that limited the impact of the program evaluation. Most of the interviewees expressed a desire for more or better communication related to other ATE projects. Having this information would allow the project to compare themselves to others and get ideas about program improvement. In the words of one PI, dissemination of this type of information is important to the project for the following reason:

> Because it would have allowed me to have a better perspective nationally rather than simply locally. I would have liked to have seen what other people are doing or at least to have had some feeling for trends and that sort of thing. (ATE Interviewee 9)

Another PI suggested that having a query option for projects to compare their own results to other specific types of projects would have been useful. This PI explained,

> If they could do anything to share any ides, it would be great versus just asking the same things over and over again in a different format….I think an ongoing newsletter and a report at a conference would be good. (ATE Interviewee 5)

The national scope of the program evaluation appeared to both help and hinder the impact. The evaluation gave some projects a broader perspective – forcing them to look beyond their own project. One PI explained:

> I think its national focus was really important. When I go in October ATE meeting, that is one of the very few opportunities we really have to be able to step beyond ourselves and the evaluations and the surveys and the other tools were really a big help in assisting us/me from being quite so provincial. (ATE Interviewee 9)

However, the broad scope necessitated more general questions, which were not easily tailored to individual projects, so projects were forced to often select "other." One PI suggested that more involvement may have addressed this short-coming.

> I think there probably should have been and could have been more involvement by some of the centers. Especially to take a look at the results and how them may implement those or make changes and how they can better use their services." (ATE Interviewee 8)

One project thought that the having online access to the annual web-based survey, along with its organization, contributed positively to the evaluation's impact. Additionally, another PI acknowledged that ATE's understanding about the time it takes to make a difference helped the program evaluation have an impact. The PI explained:

> One of the best things about the ATE program is that they tend to stick with their colleges and their PIs and their projects long enough for them to make a difference. Sticking with it long enough to get some of that data is really, really important. (ATE Interviewee 6)

Project-Level Summary

This section of the case study summarizes the ways in which ATE project leaders and evaluators felt involved in the ATE program evaluation and subsequently used the results. Although the program evaluation had several components as discussed in the "ATE Evaluation Background" section of this report, survey respondents and interviewees seem to associate the program evaluation almost exclusively with the annual web-based survey. The fact that the annual web-based survey was expected every year (after the project's first year of funding) seemed to override the other activities in which respondents may have been involved. Consequently,

there was generally a low to moderate feeling of involvement other than the time spent completing the annual survey.

However, low levels of involvement did not translate into equally low levels of use. The majority of respondents acknowledged that the program evaluation influenced their beliefs about evaluation and was, in general, useful for making project-related decisions. Despite the tedium often associated with the annual web-based survey, respondents acknowledged the importance of the evaluation to the STEM and evaluation fields in that it is a necessary step in securing additional programmatic funding.

Respondents suggested that several factors limited the impact of the ATE program evaluation, especially the lack of resources. In addition, the generality of the program evaluation, which was due to its broad, national scope, created a feeling among some projects that their project was not being represented favorably or meaningfully. Limitations aside, the collaboration fostered by the program evaluation elicited gratitude from the respondents. The evaluation also increased awareness among project leaders and evaluators about the extent to which the project was meeting its goals.

Influence on the Field

This section examines factors that appear to be related to use and influence of the ATE program evaluation on the fields of STEM education and evaluation. It presents finding resulting from four different data collection activities and is organized according to the following three topics:

- Projects' Perceptions of Influence on the Field
- The "Field's" Perceptions of Influence on the Field
- Evaluation Products' Influence on the Field

Projects' Perceptions of Influence on the Field

Our first set of findings related to influence on the field was drawn from the Project Leader and Evaluator Survey and interviews. Respondents were asked directly about their perceptions of the impact of the ATE program evaluation on STEM education and evaluation. Responses from the survey and all interviews are summarized below.

As shown below in Table 15, online survey responses to questions about the influence of the evaluation on the broad STEM education and evaluation communities were similar, with the greatest proportions of respondents stating that it was marginally to somewhat influential.

Table 15. Influence of program evaluation on the field

	Not influential at all	Marginally influential	Somewhat influential	Influential	Very influential	Total
STEM Community	21	54	55	34	11	175
	12.0%	30.9%	31.4%	19.4%	6.3%	100.0%
Evaluation Community	20	54	54	36	13	177
	11.3%	30.5%	30.5%	20.3%	7.3%	100.0%

Although the survey results indicate only the ATE program evaluation was marginally to somewhat influential on the STEM and evaluation fields, both rounds of interviews (all levels of involvement/use and the high user interviews) suggest higher levels of impact. The strongest impact of the evaluation on the broader STEM and evaluation communities was the message it sent about the importance of evaluation by placing an emphasis on accomplishments and accountability. For example, one project leader explained:

> Well, again, I think one of the key things is how important evaluation and assessment is on all of these projects. I mean basically you need to be able to do an evaluation and assessment and be able to demonstrate that yes, in fact, we're accomplishing what we said we were going to accomplish; and in fact, are we…I mean it's one thing to say well, we're going to have an impact, and we have a very strong evaluator and assessment person with our team, and the Western Michigan project just reinforces that purpose and that emphasis. (ATE Interviewee 1)

One evaluator also agreed that this evaluation sent a positive message to the STEM education community:

> I think one way there's a big impact is that it sends a message that evaluation must occur…. The fact that ATE says don't be asking for money unless you have an evaluation plan, bottom line, that changes the nature of the evaluation. (ATE Interviewee 2)

One project leader reported that the STEM education field specifically may benefit from the program evaluation because it creates methods for documenting and disseminating project activities. For example:

> …what we don't do a very good job with is documenting what the impact and outcomes of those activities are, and I think the Western Michigan project is very similar to addressing student outcomes at a university or community college. We know that we're teaching students and we know that they're very successful at becoming employed and completing the programs, but we don't do a good job of documenting that. I think what the Western Michigan project does for NSF/ATE is it provides an actual format for documenting what those outcomes and impact have not only on our local community, but also on a national basis…Again, I think it helps not only collect that information, but also disseminate that so then you can go to the brochures, with the literature on the

> website and identify what…again, to see what actual impact that had, in terms of STEM projects in education. (ATE Interviewee 1)

All but one of the high user interviewees suggested that the program evaluation impacted theSTEM field. For example, one interviewee explained that the ATE program evaluation made the evaluation process easier for those in STEM who are not professional grant writers, administrators, or evaluators. In essence, it ensured that evaluation would be embedded in STEM projects. In addition, it created a mechanism for all of the grants to report activities, and, as a result, demonstrated the impact of the ATE program to the field. One PI explained it as follows:

> I think what it's done is it's really helped to show exactly the impact that STEM has on different programs throughout the nation. I mean it got the information out there. People use it and look at it. (ATE Interviewee 7)

Regarding the evaluation field specifically, interviewees were more likely to cite general descriptions of influence than distinct ways in which the evaluation field has been influenced by the ATE program evaluation. However, one PI speculated that it could have if evaluators were "looking for guidelines and examples of a good evaluation project that will be very helpful for them." (ATE Interviewee 1)

An evaluator noted that the occasional published article may have an impact on the evaluation community:

> Well, if I consider myself part of the evaluation community, I can say in the evaluation journals that I get, every so often there is something about how you evaluate STEM programs; it tends to be written by researchers rather than STEM implementers, but I think that there is some impact, yes, on the evaluation community. (ATE Interviewee 2)

Most high user interviewees also agreed that the program evaluation had a positive impact on the evaluation field, primarily because it helped local evaluators improve their skills. In addition, it emphasized the importance of evaluation and "ensured that nobody runs these projects without evaluation being a component". One PI also explained that the importance placed on the evaluation helped "professionalize" the field.

The "Field's" Perception of Influence on the Field

The research team attempted to measure the field's perception of influence by surveying current NSF PIs and journal editors. The first of these two was the NSF PI Survey, which gathered PIs' perceptions related to evaluations, people, and publications. The researchers reasoned that if survey respondents mentioned the names and titles of the ATE program evaluation, then this might indicate the presence of a network of influence among the four targeted programs and STEM educators and evaluators. The research team examined the

survey results for evidence of influence of the ATE evaluation, the PI, and related publications on the broader fields of STEM education and evaluation.

Sixty-four of 191 respondents indicated they were able to identify an evaluation by name, but only 51 of the 64 respondents actually provided the name of an evaluation in a follow-up question. Within these 51 named evaluations, the ATE evaluation was named four times. In addition, researchers asked survey respondents to note STEM evaluation publications they viewed as influential. Four of 44 respondents named a publication associated with ATE. Finally, the team asked respondents to identify influential STEM evaluators and the evaluator associated ATE was not present in the responses.

Focusing further, there were 21 respondents who reported that they were involved in ATE. Three of these respondents identified as influential an ATE-affiliated evaluation, publication, or evaluator by name.

As a second look at the field's perception of influence on the field, the research team conducted a 3-question survey of STEM education and evaluation journal editors. Two of the three editors that responded to the journal editor survey named one STEM education evaluator who was the PI of the ATE evaluation.

Evaluation Products' Influence on the Field

Our final measure of influence on the field considers the extent to which ATE evaluation products were cited throughout the field. The citation analysis used bibliographic methods to gather data about the use of program evaluation products by individuals within the STEM education and program evaluation communities. Citation analysis consists of tracking the number of citations other authors make to a published work. This section describes both products of the ATE evaluation and the unique publications (referred to as "citing works") that cited one or more of the ATE evaluation products. Following the product and citing work descriptions, patterns in the citation data are explored through network maps.

Descriptions of the ATE Evaluation Products. The ATE evaluation produced a total of 98 evaluation related products. A list of these products and the citing works to each product is presented in Appendices A and B. The products were coded by their type, field, and subfield according to the definitions presented below:

Evaluation products were coded by type according to the following:

- Instrument/tools: evaluation instruments (observation protocols, surveys, etc), instrument manuals, online evaluation tools (logic model tool, resource database);
- Presentations: conference presentations, presentations to NSF or other audiences;
- Publications: journal articles, books, book chapters, published monographs, dissertations, newsletters;
- Reports: evaluation reports, briefing papers, fact sheets, brochures.

Product fields were broadly defined according to the following:

- STEM education/research: general topics or research related to STEM education;
- STEM evaluation: topics or findings connected to a specific STEM evaluation;
- Education—general: education areas/topics outside of STEM education;
- Evaluation—general: evaluation areas/issues outside of STEM evaluation;
- Other: items not falling into any of the above categories

Products were coded as to the specific sub-field within education or evaluation according to the following:

- Professional development: teacher training during service including the effects of professional development on student achievement, etc.;
- Workforce development: high tech, STEM workforce training/development;
- Partnerships/systemic reform: university, school, community partnerships, or state or district-wide systemic reform initiatives;
- Curriculum/instruction/assessment: related to classroom teaching (i.e., instruction methods, curriculum, pedagogy, student assessment, standards-based instruction, math/science reform);
- Evaluation: evaluation-related topics such as multi-site evaluation theory or other general evaluation theory or knowledge, including evaluation instruments, protocols, and manuals;
- Materials development: development of instructional materials;
- Project capacity building: issues related to program improvement, sustainability, advisory committee development, dissemination, replication;
- Other: other fields.

**Please note that evaluation instruments, manuals, and protocols were coded as "evaluation" rather than by the topic the instrument collected data about.*

ATE Evaluation Product Types. Over half of the products ATE produced were evaluation reports (n=52), while presentations (n=27) and publications (n=19) comprise the remaining half. ATE did not publish any evaluation instruments or tools (see Table 16).

Table 16. Types of products produced by the ATE evaluation

Product Type	N	%
Report	52	53.1
Presentation	27	27.6
Publication	19	19.4
Total	*98*	*100.0*

Significant differences were found in the mean number of citations for different types of evaluation products (F=13.902, df=2, p<.001). The differences in product type accounted for 17.1% of the variance in mean citations. As shown in Table 17, ATE publications were cited most frequently, averaging 1.26 citations per product. This was significantly higher than the number of citations for presentations or reports. It is interesting to note that none of the 27 ATE-evaluation presentations was cited.

Table 17. Citations for types of products

Evaluation Product Type	Number of products	Number of Citations	Mean Citations/ Product
Publication	19	30	1.26
Report	52	34	.38
Presentation	27	0	.00
Total	*98*	*64*	*.45*

The ATE evaluation produced a high proportion of STEM evaluation products (78%), followed by a number of general evaluation products (18%). There were also a few STEM education and research products that had a broader focus than the ATE evaluation specifically, as well as one product coded as a general education product (see Table 18).

Table 18. Fields of evaluation products

Product Fields	N	%
STEM evaluation	76	77.6
Evaluation – general	18	18.4
STEM education/research	3	3.1
Education – general	1	1.0
Total	*98*	*100.0*

Approximately one third of the ATE evaluation products were related to the topic of workforce development (n=32), not surprising given that the purpose of the ATE program is to develop a high-tech workforce. The evaluation also produced a high number of evaluation-related

products (n=38). There were also a number of project capacity building, workforce development and professional development related-products (see Table 19).

Table 19. Specific Fields of evaluation products

Product Subfields	N	%
Evaluation	38	38.8
Workforce Development	32	32.7
Project Capacity Building	14	14.3
Materials Development	6	6.1
Professional Development	5	5.1
Curriculum/Instruction/Assessment	2	2.0
Partnerships/systemic reform	1	1.0
Total	*98*	*100.0*

Significant differences were not found in the mean number of citations received by products of different fields or different subfields.

Description of the Works Citing the ATE Evaluation Products. A total of 64 references were found to the 98 evaluation products produced by the ATE evaluation. The 64 references came from 42 unique citing works, which are detailed in Appendix B. Table 20 shows that although the vast majority of the citing works cited only one product (n=29), eight of the citing works cited two evaluation products, and the remaining five cited more than three products. All five of the citing works that cited three or more ATE products were themselves other ATE products.

Table 20. Number of references to evaluation products by citing works

Number of references per citing work	N	%
1	29	69.0
2	8	19.0
3	2	4.8
4	2	4.8
5	1	2.4
Total	*42*	*100.0*

As shown in Table 21, the greatest proportion of citations came from sources external to the ATE evaluation (45%), meaning that they were not affiliated with the other three evaluations examined in the Beyond Evaluation Use project, those evaluations' PIs, or other NSF-related sources. After external sources, the second most common source of citations was from the ATE

evaluation itself (43%). In many cases, these were references from one evaluation report to previously published evaluation reports, as is a common technique for highlighting work done in the past without having to repeat the details. There were also three citations from ATE projects or centers, as well as one from an NSF source, and one from Frances Lawrenz, the PI of the CETP program, although the citation was not from a CETP publication.

Table 21. Authors of citing works

Citing Work Authors	N	%
Other	19	45.2
ATE Evaluation	18	42.9
ATE Project/Center	3	7.1
NSF-related	1	2.4
PI	1	2.4
Total	*42*	*100.0*

The citing works were predominantly US in origin, in terms of where they were published or presented. However, six of the forty-two citations came from international sources: three from China, one from Spain, one from Sweden, and one from Thailand.

Citing works were categorized by type according to the same categorization scheme used with the products, with the addition of codes for "electronic resources" which was used for Web pages, archived listserv messages, or similar online sources, and the category "other." As shown in Table 22, publications were the most common source for citations to ATE products (43%), followed by evaluation reports (31%), and to a lesser extent presentations (12%) and electronic resources (12%). One citation came from a course syllabus.

Table 22. Citing work types

Types	N	%
Publication	18	42.9
Report	13	31.0
Presentation	5	11.9
Electronic Resource	5	11.9
Other	1	2.4
Total	*42*	*100.0*

As shown in Table 23, the majority of the citing works came from within the STEM fields with STEM evaluation-related works (38%) and STEM education and research-related works (31%) predominating. A sizeable proportion of citations came from evaluation publications that

addressed general evaluation issues (21%). General education publications and other sources were less common.

Table 23. Citing work fields

Fields	N	%
STEM evaluation	16	38.1
STEM education/research	13	31.0
Evaluation	9	21.4
Education – general	2	4.8
Other	2	4.8
Total	*42*	*100.0*

The citing works represented a variety of specific fields within education and evaluation. As shown in Table 24, there were equal numbers of citing works that addressed issues in workforce development (n=12) and evaluation (n=12), while a number of other citing works were related to project capacity building, materials development, and partnerships. There were also two publications from the area of professional development, one from curriculum and instruction, one from medicine, and one from wellness/fitness programs.

Table 24. Citing work sub-fields

Subfields	N	%
Workforce Development	12	28.6
Evaluation	12	28.6
Project Capacity Building	6	14.3
Materials Development	4	9.5
Partnerships/systemic reform	3	7.1
Professional Development	2	4.8
Other	2	4.8
Curriculum/Instruction/Assessment	1	2.4
Total	*42*	*100.0*

Citation Patterns of ATE Core Evaluation Products. Although the descriptions presented above convey one picture of the citations to ATE products, this section presents a graphical picture of the interconnected web of ATE citations. For the following images, the program NetDraw was used to create maps of the networks of ATE products and citations. In each of the maps, arrows point from the citing work to the evaluation product it is citing. The products and citing works are in the same position in each of the maps; only the colors change to highlight different characteristics of interest to help describe citation patterns.

Figure 1 presents the overall citation structure of ATE products (blue squares) and works authored by individuals not affiliated with the ATE evaluation (red circles). As shown on the left side of the figure, there are a number of products that were cited by only one or two internal or external works. On the right side of the figure, there is a mass of ATE products that refer to each other, along with one external citation. Six of the products in this mass were published part of the "Advanced Technological Education Program Evaluation Briefing Paper Series" and, as shown, commonly one paper in the series cited one or more previously published papers in the series. The external citation was from an ATE Centers Impact report produced by an individual not affiliated with ATE program evaluation.

In the center of the figure, there are a number of ATE products that are linked together. Many of the products in this citation chain were articles published together in the "The ATE Program: Issues for Consideration" monograph. The works in the monograph were cited by both internal and external sources in subsequent years. In general, the citation patterns presented below highlight that the citations to ATE products were primarily self-citations from other ATE products, although a number were from outside sources. This demonstrates that the evaluation products have greater influence on work within the evaluation, rather than having a large amount of documentable influence on external publications.

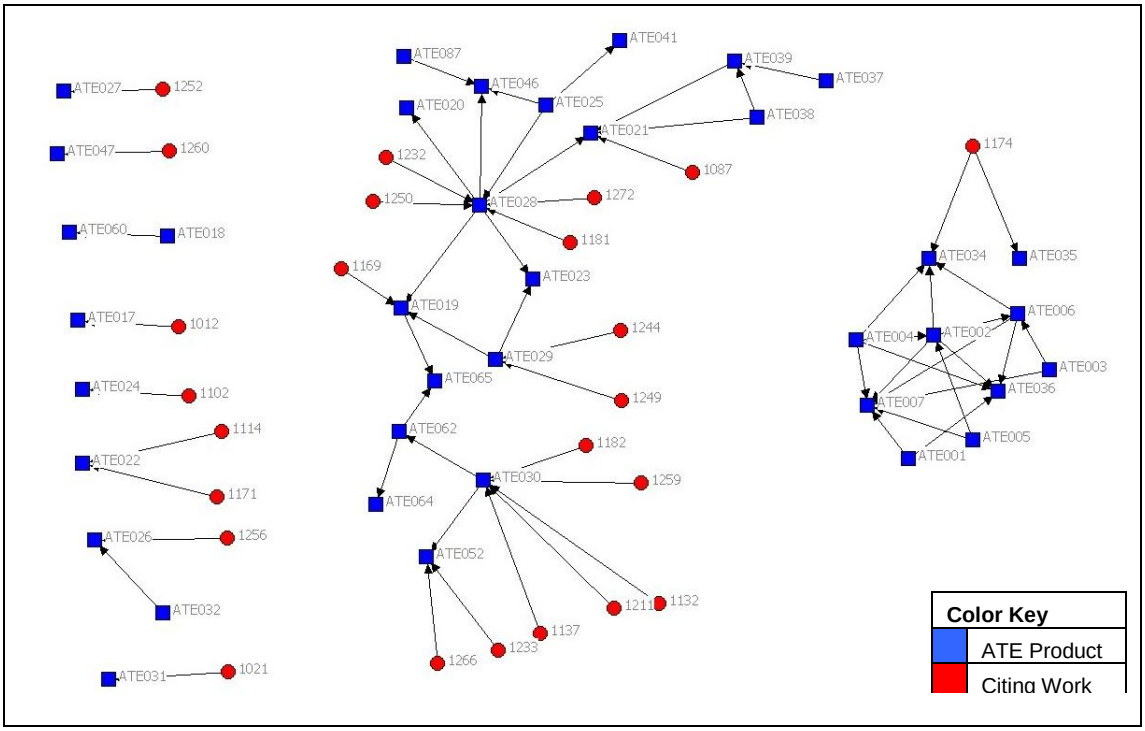


Figure 1. ATE products and citations

In Figure 2, the color coding scheme is changed to illustrate the different authors of the external citing works. ATE products are again represented as blue squares. The citing works are colored as to whether they were affiliated with one of the other three program evaluations

under consideration in this study (i.e., CETP, LSC, and MSP-RETA), other NSF sources, or from sources not affiliated with NSF or any of the program evaluations (see color key).

A number of the citing works to ATE products were by authors not affiliated with the four program evaluations or NSF (represented by red circles). The works from these non-affiliated sources primarily cited five ATE publications:

- ATE022: Owens, T. (2002). Dissemination: A key element of the ATE program. In A. R. Gullickson, F. P. Lawrenz, & N. Keiser (Eds.), *The ATE program: Issues for consideration: A monograph* (pp.26-44). Washington, DC: National Science Foundation.
- ATE028: Zinser, R., & Lawrenz, F. P. (2004). New roles to meet industry needs: A look at the Advanced Technological Education program. *Journal of Vocational Education Research, 29*(2), 85-100.
- ATE029: Lawrenz, F. P., Keiser, N., & LaVoie, B. (2003). Sustaining innovation in technological education. *Community College Review, 30*(4), 1-14.
- ATE030: Lawrenz, F. P., Keiser, N.,& LaVoie, B. (2003). Evaluative site visits: A methodological review. *American Journal of Evaluation, 24*(3), 341-352.
- ATE052: Lawrenz, F. P., Keiser, N., & Lavoie, B. (2002). *A guide for planning and implementing site visits.* Kalamazoo, MI: Western Michigan University: The Evaluation Center.

The citing works were all publications or reports that addressed issues in project capacity building, workforce development, or evaluation.

Additionally, Figure 2 shows that the authorship of three citing works (yellow circles) was attributed to ATE projects; one attributed authorship to a CETP project (green circle), and one was authored by an NSF source (orange circle). Each of these authors was drawn to different ATE products, interestingly.

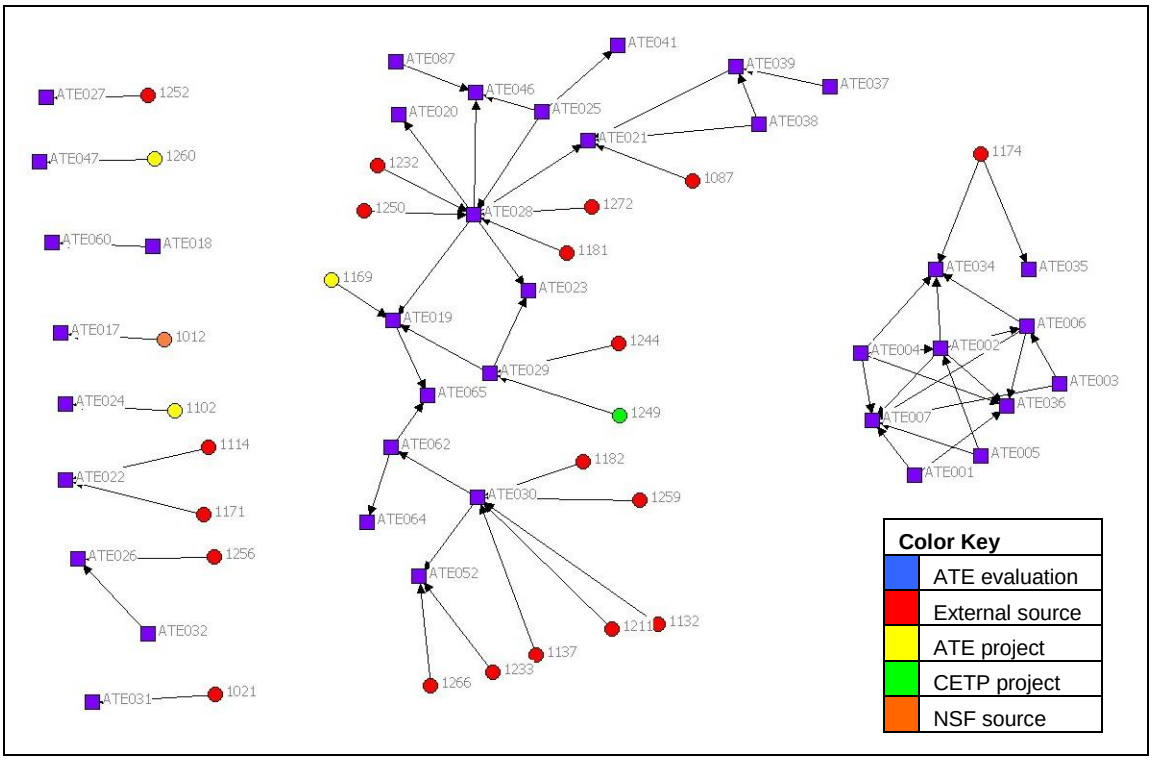


Figure 2. Citing authors of ATE products

In Figure 3, the products and citations are all re-colored to represent the specific fields that the publication addressed; a color key is provided with the figure. The ATE products are squares and other works are circles.

As shown in the figure, the ATE products and citing works covered a wide variety of fields, and there appears to be a link between the fields of the products and their citations. For example, products ATE030 [Lawrenz, F. P., Keiser, N,.& LaVoie, B. (2003). Evaluative site visits: A methodological review.] and ATE052 [Lawrenz, F. P., Keiser, N., & Lavoie, B. (2002). *A guide for planning and implementing site visits.*] were evaluation-related products, which were cited predominantly by evaluation-related citing works. Similarly, ATE028 [Zinser, R., & Lawrenz, F. P. (2004). New roles to meet industry needs: A look at the Advanced Technological Education program.] addresses issues in workforce development and many of its citing works are also related to workforce development issues.

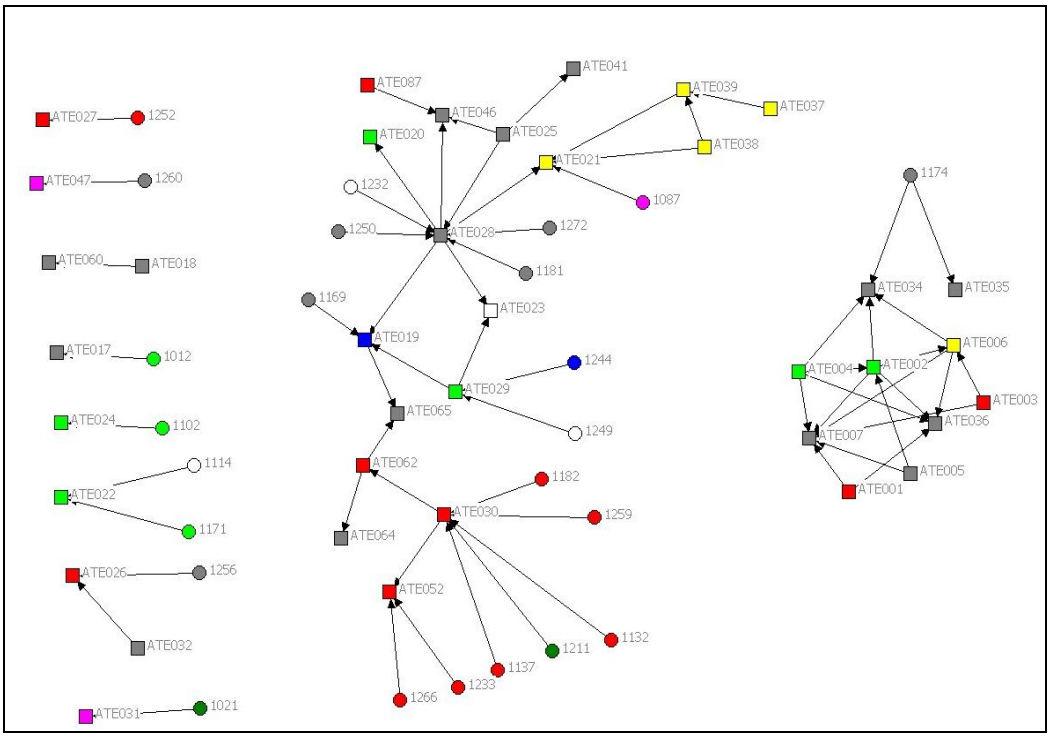


Figure 3. Fields of citing works and products

Citation Analysis Discussion and Conclusions. The citation patterns described above suggest that the products produced by the ATE program evaluation were primarily influential on other ATE sources. This does not mean, however, that the knowledge produced by the ATE evaluation was limited in its influence to a small group of evaluators engaged in its day to day work. In fact, there were 24 separate individuals who were listed as authors on one or more of the ATE evaluation products, as the ATE evaluation engaged a variety of internal staff and external consultants in its work. That said, these citation levels and patterns do suggest that the greatest influence of the ATE evaluation products was on those affiliated with the ATE national evaluation in some way, as opposed to the field more broadly.

As far as influence on the fields of STEM education and evaluation, it appears that there were a few key evaluation-related publications and workforce development publications that had the greatest influence on individuals who were not affiliated with the ATE evaluation in any direct way. The ATE evaluations' published articles and monograph were most often used externally, whereas reports were primarily cited by internal sources, and presentations were not cited in any other works.

Influence on the Field Summary

This research study employed four distinct measures to assess the ways and extent to which the ATE program influenced the field of STEM education and evaluation. Although the influence may not have been great – given that a majority of survey respondents indicated the program evaluation was "marginally influential" or "somewhat influential" on the field and the

NSF PI Survey results showed limited influence – the ATE did have some notable influences on the fields of STEM education and evaluation. Specifically, at least at least a quarter of the ATE project leaders and evaluators indicated a high level of influence and interviewees described both general and specific examples of impact.

Findings suggest that evaluation publications may positively affect the influence on the field. The more specific examples from the interviews related to influence on the field corroborate findings from the citation analysis and the editor inquiry. The citation analysis suggests that a few key evaluation-related publications and workforce development publications had an influence on individuals who were not affiliated with the ATE evaluation in any direct way. Results from the citation analysis also showed that the ATE evaluations' published articles and monograph were most often used externally than internally. Although the editor inquiry was small with only three editors, two of the three mentioned the ATE Program PI as a STEM education evaluator. One can assume that it was ATE-related publications that heightened the editors' awareness.

In conclusion, the ATE program evaluation, especially its publications, appeared to influence the field, albeit perhaps not to a great extent. Whether it was a detailed interview example, a citation in a publication, a response under "influential evaluation" by an NSF PI, or being noted by a journal editor, in every attempt to measure influence, at least a small amount was found.

Overall Summary: Relationship between Involvement and Use

The ATE program evaluation focused on program-wide productivity, with the primary audience being NSF. Evaluators structured the evaluation in a manner that would allow them to provide feedback to NSF, so that NSF could use the evaluation information for programmatic decisions and accountability reports to Congress. Consequently, the evaluation was not designed to necessarily influence the projects and centers. With that said, program-level feedback was regularly disseminated to projects in hopes of improving project-level processes and outcomes. Principal evaluation activities included: collecting descriptive data, developing an annual web-based evaluation survey to collect data over time, conducting site visits, drafting targeted reports on various aspects of the program, and providing formative evaluation information to NSF and to the projects and centers.

In terms of involvement in the evaluation process, the process of creating the survey framework and the question wording involved NSF program officers and an evaluation advisory group. The evaluation advisory group, comprised of individuals who worked directly with ATE projects and centers, helped conceptualize the survey and contributed to item development. Individual ATE grantees had relatively little participation or input primarily because the evaluation focused more on NSF's needs, not on needs of individual projects.

This case study showed that, overall, involvement in the program evaluation was generally low to moderate, centering primarily on the annual web-based survey that was expected of all

projects by their second year of funding. However, the reported levels of involvement did not translate into equally low levels of use. In general, results from the Project Leader and Evaluator survey and the interviews suggest that the evaluation did influence project leaders and evaluators; in some cases extensively. However higher levels of influence were not directly related to higher levels of involvement in the evaluation. Other variables that may have affected use included project perceptions related to the extent to which the evaluation was representative of their own project and the opportunities the program provide for interaction and collaboration with other projects and evaluators. Influence on the field was less apparent than the influence of the projects, but still notable. Publications related to the evaluation show the strongest potential for influencing the field.

In summary, in the case of the ATE program evaluation the relationship between involvement and use is weak. However, the strength of the relationship is more a result of few opportunities for involvement than of instances of use or influence. This case study presented evidence of at least some influence of the program evaluation on both the projects and the field, despite lower levels of involvement.

Appendix A: ATE Evaluation Products

Product ID	Product Reference Information	Product Type	Product Field	Product Subfield	# Cites	Citation IDs
ATE001	Gullickson, A. R., Coryn, C. L. S., & Ritchie, L. A. (2006). Program evaluation. [aka "Project level evaluation practices"] [Advanced Technological Education Program Evaluation Briefing Paper Series, Briefing Paper #1]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Evaluation	0	
ATE002	Coryn, C. L. S., Gullickson, A. R., & Ritchie, L. A. (2006). Contributors and inhibitors influencing program improvement [Advanced Technological Education Program Evaluation Briefing Paper Series, Briefing Paper #2]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Project Capacity Building	2	ATE005, ATE004
ATE003	Ritchie, L. A., Gullickson, A. R., & Coryn, C. L. S. (2006). Workforce needs assessment [Advanced Technological Education Program Evaluation Briefing Paper Series, Briefing Paper #3]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Evaluation	0	
ATE004	Coryn, C. L. S., Gullickson, A. R., & Ritchie, L. A. (2006). Contributors and inhibitors influencing articulation agreements [Advanced Technological Education Program Evaluation Briefing Paper Series, Briefing Paper #4]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Project Capacity Building	0	
ATE005	Ritchie, L. A., Gullickson, A. R., & Coryn, C. L. S. (2006). Challenges and resolutions [Advanced Technological Education Program Evaluation Briefing Paper Series, Briefing Paper #5]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	0	
ATE006	Gullickson, A. R., Coryn, C. L. S., & Ritchie, L. A. (2006). Materials development, professional development, and program improvement: Productivity and quality [Advanced Technological Education Program Evaluation Briefing Paper Series, Briefing Paper #6]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Materials Development	2	ATE002, ATE003

ATE007	Coryn, C. L. S., Ritchie, L. A., & Gullickson, A. R. (2006). Advanced Technological Education Program Fact Sheet. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	6	ATE002, ATE006, ATE001, ATE005, ATE003, ATE004
ATE008	The Evaluation Center (2005). 2005 ATE survey fact sheet (Centers). Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	0	
ATE009	The Evaluation Center (2005). 2005 ATE survey fact sheet (Projects and articulation partnerships). Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	0	
ATE010	The Evaluation Center (2005). 2005 ATE survey fact sheet. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	0	
ATE011	The Evaluation Center (2004). 2004 ATE survey fact sheet. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	0	
ATE012	The Evaluation Center (2003). 2003 ATE survey fact sheet. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	0	
ATE013	Gullickson, A. R., & Hanssen, C. E. (2006). Local evaluation in multisite STEM programs: relating evaluation use and program results. In D. Huffman & F.L. Lawrenz (Eds.), New Perspectives on the Evaluation of Science, Technology, Engineering and Mathematics, New Directions in Evaluation.	Publication	STEM evaluation	Evaluation	0	
ATE014	Welch, W and Lawrenz, F. (2004). The use of evaluation in the advanced technological education program. In Evaluating the upgrading of technical courses at two year colleges: NSF's advanced technological education program. Advances in Program Evaluation Volume 9 Elsevier: London.	Publication	STEM evaluation	Evaluation	0	
ATE015	Welch, W. & Gullickson, A. (2004). The role of advisory committees. In Evaluating the upgrading of technical courses at two year colleges: NSF's advanced technological education program. Advances in Program Evaluation Volume 9 Elsevier: London.	Publication	STEM evaluation	Project Capacity Building	0	

ATE016	Gullickson, A. R., Lawrenz, F. P., & Keiser, N. (Eds.). (2004). Evaluating the upgrading of technical courses at two year colleges: NSF's advanced technological education program. Advances in program evaluation, 9. London, UK: Elsevier.	Publication	STEM evaluation	Evaluation	0	
ATE017	Gullickson, A. R., Lawrenz, F. P. & Keiser, N. (Eds.). (2002). The ATE program: Issues for consideration: A monograph. Washington, DC: National Science Foundation.	Publication	STEM evaluation	Workforce Development	1	1012
ATE018	Gullickson, A. R., & Tressler, G. R. (2002). Recruitment and retention. In A. R. Gullickson, F. P. Lawrenz, & N. Keiser (Eds.), The ATE program: Issues for consideration: A monograph (pp. 103-129). Washington, DC: National Science Foundation.	Publication	STEM evaluation	Workforce Development	0	
ATE019	Gold, N., & Powe, K. W. (2002). Assessing the impact and effectiveness of professional development. In A. R. Gullickson, F. P. Lawrenz, & N. Keiser (Eds.), The ATE program: Issues for consideration: A monograph (pp. 81-102). Washington, DC: National Science Foundation.	Publication	STEM evaluation	Professional Development	3	ATE029, 1169, ATE028
ATE020	Reed, L. W. (2002). Program improvement efforts: Approaches and results in improving technical education programs by ATE centers and projects. In A. R. Gullickson, F. P. Lawrenz, & N. Keiser (Eds.), The ATE program: Issues for consideration: A monograph (pp. 62-80). Washington, DC: National Science Foundation.	Publication	STEM evaluation	Project Capacity Building	1	ATE028
ATE021	Rogers, G. R. (2002). Materials development and the ATE program. In A. R. Gullickson, F. P. Lawrenz, & N. Keiser (Eds.), The ATE program: Issues for consideration: A monograph (pp. 45-61). Washington, DC: National Science Foundation.	Publication	STEM evaluation	Materials Development	4	ATE038, 1087, ATE039, ATE028
ATE022	Owens, T. (2002). Dissemination: A key element of the ATE program. In A. R. Gullickson, F. P. Lawrenz, & N. Keiser (Eds.), The ATE program: Issues for consideration: A monograph (pp.26-44). Washington, DC: National Science Foundation.	Publication	STEM evaluation	Project Capacity Building	2	1114, 1171
ATE023	Reed, L. W. (2002). Collaboration efforts: Approaches, uses, and effectiveness of collaboration. In A. R. Gullickson, F. P. Lawrenz, & N. Keiser (Eds.), The ATE program: Issues for consideration: A monograph (pp. 10-25). Washington, DC: National Science Foundation.	Publication	STEM evaluation	Partnerships/ systemic reform	2	ATE029, ATE028

ATE024	Lawrenz, F. P., & Keiser, N. (2002). Sustainability: Increasing the likelihood of a long-term impact by the ATE program. In A. R. Gullickson, F. P. Lawrenz, & N. Keiser (Eds.), The ATE program: Issues for consideration: A monograph (pp. 130-146). Washington, DC: National Science Foundation.	Publication	STEM evaluation	Project Capacity Building	1	1102
ATE025	Zinser, R., & Hanssen, C. E. (2006). Improving access to the baccalaureate: Articulation Agreements and the National Science Foundation's Advanced Technological Education Program. Community College Review. Vol. 34, No. 1, 27-43	Publication	STEM evaluation	Workforce Development	0	
ATE026	Keiser, J., Lawrenz, F. P., & Appleton, J. (2004). Technical education curriculum assessment. Journal of Vocational Education Research, 29(3), 181-194.	Publication	STEM education/ research	Evaluation	2	ATE032, 1256
ATE027	Hanssen, C. E. (2004). Structural equation modeling as a tool for multisite evaluation. Dissertation Abstracts International, 65(04), 1224A. (UMI No. 3130605).	Publication	Evaluation - general	Evaluation	1	1252
ATE028	Zinser, R., & Lawrenz, F. P. (2004). New roles to meet industry needs: A look at the Advanced Technological Education program. Journal of Vocational Education Research, 29(2), 85-100.	Publication	STEM evaluation	Workforce Development	5	1181, 1232, 1250, ATE025, 1272
ATE029	Lawrenz, F. P., Keiser, N., & LaVoie, B. (2003). Sustaining innovation in technological education. Community College Review, 30(4), 1-14.	Publication	STEM education/ research	Project Capacity Building	2	1244, 1249
ATE030	Lawrenz, F. P., Keiser, N,.& LaVoie, B. (2003) Evaluative site visits: A methodological review. American Journal of Evaluation, 24(3), 341-352.	Publication	Evaluation - general	Evaluation	5	1132, 1137, 1182, 1211, 1259
ATE031	LaVoie, B. (2003). The development and validation of the biotechnology problem-solving skills assessment for community college biotechnology students. Doctoral Dissertation: University of Minnesota.	Publication	STEM evaluation	Curriculum/ Instruction/ Assessment	1	1021
ATE032	Gullickson, A. R., Wingate, L., Lawrenz, F. A., & Coryn, C. L. S. (2006). The National Science Foundation's Advanced Technology Education Program: Final evaluation report. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	0	
ATE033	Welch, W. W., & Gullickson, A. R. (2006). The Sustainability of Advanced Technological Education Supported Efforts: An Evaluation. Kalamazoo: Western Michigan University, The	Report	STEM evaluation	Project Capacity Building	0	

	Evaluation Center.					
ATE034	Gullickson, A. R., Coryn, C. L. S., & Hanssen, C. E. (2006). ATE indicators of productivity: Six-year trends 2000-2005. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	4	ATE002, ATE006, 1174, ATE004
ATE035	Germuth, A. A., Gullickson, A. R., Lawrenz, F. P., & Hanssen, C. E. (2006). Assessing the value added by NSF's ATE program: Business and industry perspective's cross-site analysis report. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	1	1174
ATE036	Coryn, C. L. S., & Hanssen, C. E. (2005). 2005 ATE technical report: Processes, procedures, and results. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	4	ATE002, ATE006, ATE001, ATE004
ATE037	Appleton, J. J., & Lawrenz, F. P. (2005). ATE materials outcome comparison: Technical report. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Materials Development	0	
ATE038	Appleton, J. J., & Lawrenz, F. P. (2004). ATE materials development process report. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Materials Development	0	
ATE039	Keiser, J., Lawrenz, F. P., & Appleton, J. J. (2004). Evaluation of materials produced by the ATE program. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Materials Development	2	ATE038, ATE037
ATE040	Coryn, C. L. S., Gullickson, A. R., & Hanssen, C. E. (2004). Assessing the impact and effectiveness of the Advanced Technological Education (ATE) program: 2004 survey results: Volume II status of the ATE centers. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	0	
ATE041	Coryn, C. L. S., Gullickson, A. R., & Hanssen, C. E. (2004). Assessing the impact and effectiveness of the Advanced Technological Education (ATE) program: 2004 survey results: Volume III status of the ATE projects. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	1	ATE025
ATE042	Jinkerson, D., & Hanssen, C. E. (2004).Evaluation of professional development efforts of ATE projects and centers: Phase I report. Kalamazoo, MI: Western Michigan University,	Report	STEM evaluation	Professional Development	0	

	The Evaluation Center.					
ATE043	Jinkerson, D. & Hanssen, C. E. (2004). Evaluation of professional development efforts of ATE projects and centers: Phase II report. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Professional Development	0	
ATE044	Jinkerson, D., & Hanssen, C. E. (2004). Evaluation of professional development efforts of ATE projects and centers: Phase III report. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Professional Development	0	
ATE045	Jinkerson, D. & Hanssen, C. E. (2004). Evaluation of professional development efforts of ATE projects and centers: Phase IV report. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Professional Development	0	
ATE046	Hanssen, C. E., Gullickson, A. R., & Lawrenz, F. P. (2003). Assessing the impact and effectiveness of the Advanced Technological Education (ATE) program: Survey 2003: ATE program status and trends. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	3	ATE087, ATE025, ATE028
ATE047	Lavoie, B. (2003).Summary report for the development and validation of the biotechnology problem-solving skills assessment. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Curriculum/ Instruction/ Assessment	1	1260
ATE048	Welch, W., & Reineke, R. A. (2003). National Visiting Committee (NVC) handbook. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Evaluation	0	
ATE049	Pucel, D. J. (2003). Examination of the annual survey report for 2002 and suggestions for improvement in relation to program goals, and review of survey 2002 and survey 2003 in relation to the 2002 report. New Brighton, MN: Performance Training Systems, Inc.	Report	STEM evaluation	Evaluation	0	
ATE050	Rogers, G. (2002). Developing high quality materials [brochure]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Materials Development	0	

ATE051	Lawrenz, F. P., & Keiser, N. (2002).Guide for improving sustainability [brochure]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Project Capacity Building	0	
ATE052	Lawrenz, F. P., Keiser, N., & Lavoie, B. (2002). A guide for planning and implementing site visits. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	Evaluation - general	Evaluation	3	ATE030, 1233, 1266
ATE053	Reineke, R. (2002). Meeting reminders for advisory committee members [brochure]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Project Capacity Building	0	
ATE054	Welch, W. (2002). Advisory committee checklist [brochure]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Project Capacity Building	0	
ATE055	National Visiting Committee (2002). Report for year 2. Daytona, FL: iTec.	Report	STEM evaluation	Workforce Development	0	
ATE056	Gullickson, A. R., & Tressler, G. R. (2002). Best practices in recruitment and retention [brochure]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	Education - general	Workforce Development	0	
ATE057	Welch, W., & Reineke, R. A. (2002). A study of advisory committees used by ATE projects and centers: Final report (Part A). Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Project Capacity Building	0	
ATE058	Welch, W., & Reineke, R. A. (2002). A study of advisory committees used by ATE projects and centers: Final report (Part B). Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Project Capacity Building	0	
ATE059	Gullickson, A. R. Lawrenz, F. P., & Keiser, N. (2002). Survey 2002: The status of ATE projects and centers. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	0	
ATE060	Welch, W., & Gullickson, A. (2001, March). ATE evaluation project final site visit report: Advancing careers in technology and science: Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	1	ATE018
ATE061	Gullickson, A. R. Lawrenz, F. P., & Keiser, N. (2001). Survey 2001: The status of ATE projects and centers. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	0	

ATE062	Migotsky, C., & Stake, R. (2001). An evaluation of an evaluation: CIRCE's metaevaluation of the ATE program evaluation. Urbana-Champaign, IL: University of Illinois at Urbana-Champaign.	Report	STEM evaluation	Evaluation	1	ATE030
ATE063	Hartmann, D., & Loizides, G. (2001). Metaevaluation of the web-based ATE survey evaluation system. Kalamazoo, MI: Kercher Center of Social Research.	Report	STEM evaluation	Evaluation	0	
ATE064	Gullickson, A. R., Lawrenz, F. P., & Keiser, N. (2000). Status report I: The nature of the ATE program. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	1	ATE062
ATE065	Gullickson, A. R., Lawrenz, F. P., & Keiser, N. (2000). Status report II: Findings from a survey of ATE projects and centers. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	2	ATE019, ATE062
ATE066	Site Visit Evaluation Report for the Bio Link Advanced Technology Center. Lawrenz, F., Gloria Rogers and Nanette Keiser, Report available from Bio Link Advanced Technology Center, February, 2001.	Report	STEM evaluation	Workforce Development	0	
ATE067	Site Visit Evaluation Report for the Northwest Center for Sustainable Resources. Lawrenz, F., Gloria Rogers and Leonard Sterry, Report available from the Northwest Center for Sustainable Resources, November, 2000.	Report	STEM evaluation	Workforce Development	0	
ATE068	Site Visit Evaluation Report for the Maricopa Advanced Technology Center. Lawrenz, F, Nanette Keiser and Lester Reed, Report available from Maricopa Advanced Technology Center, October, 2000.	Report	STEM evaluation	Workforce Development	0	
ATE069	Site Visit Evaluation Report for the GIS/GPS Laboratory Exercises Using Workplace Data Sets at Houston Community College (SW Campus) Project, Lawrenz, F., Nanette Keiser, Report available from Houston Community College (SW Campus), September, 2000	Report	STEM evaluation	Workforce Development	0	
ATE070	Evaluation of the NSF Advanced Technology Education Program Review Panel Meeting, Report available from National Science Foundation, December, 1999.	Report	STEM evaluation	Workforce Development	0	
ATE071	Lawrenz, F., Gullickson, A. and Barbour, S. Responses to NSF's Advanced Technology Education Program Status Survey, National Science Foundation, 1999.	Report	STEM evaluation	Workforce Development	0	

ATE072	Achieving Sustainability in ATE Projects, invited address, for the National Science Foundation at the National ATE PI meeting, Washington, DC. October, 200l	Presentation	STEM evaluation	Project Capacity Building	0	
ATE073	ATE Program Evaluation invited session Frances Lawrenz, Arlen Gullickson and Nanette Keiser for the National Science Foundation at the National ATE PI meeting, Washington, DC. October, 2001	Presentation	STEM evaluation	Evaluation	0	
ATE074	ATE Program Evaluation, Frances Lawrenz, Arlen Gullickson at NSF Advanced Technology Education national PI Conference October, 2000	Presentation	STEM evaluation	Evaluation	0	
ATE075	ATE Program Evaluation. Frances Lawrenz, Arlen Gullickson and Nanette Keiser at the NSF Advanced Technology Education Program national meeting Washington DC, October, 2002	Presentation	STEM evaluation	Evaluation	0	
ATE076	ATE Program Evaluation. Frances Lawrenz, Arlen Gullickson and Carl Hanssen at the NSF Advanced Technology Education Program national meeting Washington DC, October, 2003	Presentation	STEM evaluation	Evaluation	0	
ATE077	Coryn, C. L. S., Hanssen, C. E., Gullickson, A. R., & Ritchie, L. A. (2005). Identifying relevant evaluative criteria in the multi-site context: A conceptual approach for diverse needs, values, and goals. Paper presented at the co-sponsored American Evaluation Association/Canadian Evaluation Society conference. Toronto, Ontario, Canada. October.	Presentation	Evaluation - general	Evaluation	0	
ATE078	Demonstration session How to Conduct Site Visits. Frances Lawrenz, Nanette Keiser, at American Evaluation Association national meeting Washington, DC November, 2002	Presentation	Evaluation - general	Evaluation	0	
ATE079	Development of a Web Based Evaluation System for the Advanced Technology Education Program Frances Lawrenz, Arlen Gullickson at American Evaluation Association national meeting in Hawaii, November, 2000	Presentation	STEM evaluation	Evaluation	0	
ATE080	Dissemination of Evaluation Findings and Recommendations: Emerging Approaches. Frances Lawrenz, Arlen Gullickson and Nanette Keiser at American Evaluation Association national meeting Washington, DC November, 2002	Presentation	Evaluation - general	Evaluation	0	

ATE081	Frances Lawrenz & Arlen Gullickson, (2006). The Effect of Involvement and Evaluation Procedures on National Science Foundation Policy about Advanced Technician Education. Paper presented at the European Evaluation Society international meeting. London, England.	Presentation	STEM evaluation	Evaluation	0	
ATE082	Germuth, A., & Heinemeier, S. (2004). Defining value-added: Definitional constraints and impact on interpreting evaluation findings. Paper presented at the annual meeting of the American Evaluation Association, Atlanta, GA. November.	Presentation	Evaluation - general	Evaluation	0	
ATE083	Gullickson, A. G. (2003). Interactive video conferencing systems: An evaluation point of view. Paper presented at the annual meeting of the American Evaluation Association, Reno, NV. November.	Presentation	Evaluation - general	Evaluation	0	
ATE084	Gullickson, A. R., & Hanssen, C. E. (2004). Local evaluation in multisite programs: Does a relationship exist between evaluation use and program results? Paper presented at the annual meeting of the American Evaluation Association, Atlanta, GA. November.	Presentation	Evaluation - general	Evaluation	0	
ATE085	Hanssen, C. E. (2004). Using structural equation modeling to support formative evaluation of multisite programs. Paper presented at the annual meeting of the American Evaluation Association, Atlanta, GA. November.	Presentation	Evaluation - general	Evaluation	0	
ATE086	Hanssen, C. E., & Lawrenz, F. P. (2005). Web-based surveys in multi-site program evaluation: Strengths and limitations of two National Science Foundation cases. Paper presented at the co-sponsored American Evaluation Association/Canadian Evaluation Society conference. Toronto, Ontario, Canada. October.	Presentation	STEM evaluation	Evaluation	0	
ATE087	Hanssen, C., & Coryn, C. L. S. (2004). Theory-driven evaluations of multi-site programs. Paper presented at the annual American Evaluation Association conference. Atlanta, GA. November.	Presentation	Evaluation - general	Evaluation	0	
ATE088	How Evaluation Can Be Used to Aid in Program Sustainability: Lessons Learned Based on Quantitative and Qualitative Data. Frances Lawrenz, Arlen Gullickson and Nanette Keiser at the American Evaluation Association National Meeting St. Louis, November, 2001	Presentation	Evaluation - general	Evaluation	0	

ATE089	Issue-Oriented Papers: An Approach that Encourages Diversities of Opinions about Research and Evaluation Data for NSF's Advanced Technological Education Program. Frances Lawrenz, others in a panel at American Educational Researchers Association national meeting New Orleans, April, 2002	Presentation	STEM evaluation	Evaluation	0	
ATE090	Jinkerson, D. L., & Hanssen, C. E. (2004). Methodological issues in conducting an evaluation of professional development . Paper presented at the Annual Meeting of the Canadian Evaluation Society, Saskatoon, SA. May.	Presentation	Evaluation - general	Evaluation	0	
ATE091	Jinkerson, D. L., & Hanssen, C. E. (2005). The impact and effectiveness of the Advanced Technological Education program's professional development efforts: Using a targeted-study to identify evaluative criteria. Paper presented at the co-sponsored American Evaluation Association/Canadian Evaluation Society conference. Toronto, Ontario, Canada. October.	Presentation	STEM evaluation	Evaluation	0	
ATE092	Lawrenz, F. P. (2004). Evaluation of the quality of materials developed across multiple sites and content areas. Paper presented at the annual meeting of the American Evaluation Association, Atlanta, GA. November.	Presentation	Evaluation - general	Evaluation	0	
ATE093	Lawrenz, F. P., & Hanssen, C. E. (2003). A toolkit building workshop on site visits. Session conducted at the annual meeting of the American Evaluation Association, Reno, NV. November.	Presentation	Evaluation - general	Evaluation	0	
ATE094	Lawrenz, F. P., & Keiser, J. (2004). New directions in vocational education research. Paper presented at the annual meeting of the American Educational Research Association, San Diego, CA. April.	Presentation	STEM evaluation	Evaluation	0	
ATE095	Lessons Learned from the Development and Use of a Web-based Survey to Collect Evaluation Data. Frances Lawrenz, Arlen Gullickson and Nanette Keiser at the American Evaluation Association National Meeting St. Louis, November, 2001	Presentation	Evaluation - general	Evaluation	0	

ATE096	Schroeter, D. C., & Morell, J. (2005). Criteria for evaluating program sustainability within a multi-site context: A longitudinal approach. Paper presented at the co-sponsored American Evaluation Association/Canadian Evaluation Society conference. Toronto, Ontario, Canada. October.	Presentation	Evaluation - general	Evaluation	0	
ATE097	The Effective Use of Site Visits in the Evaluation of a National Program. Frances Lawrenz, Arlen Gullickson and Nanette Keiser at the American Evaluation Association National Meeting St. Louis, November, 2001	Presentation	Evaluation - general	Evaluation	0	
ATE098	The Effectiveness of NSF's Move into Technician Education at Community Colleges, at American Educational Research Association national meeting in New Orleans, April, 2000	Presentation	STEM education/ research	Workforce Development	0	

Appendix B: Citing Works to ATE Evaluation Products

Citation ID	Citing Work Reference	Citation Type	Citation Field	Citation Subfield
1012	Bailey, Thomas R.; Matsuzuka, Yukari; Jacobs, James; Morest, Vanessa Smith; Hughes, Katherine L. (2003). Institutionalization and Sustainability of the National Science Foundation's Advanced Technological Education Program. ERIC document: ED482183.	Report	STEM evaluation	Project Capacity Building
1021	China Medical College collection number: 743. http://211.20.11.205:8080/edissstat/customPDFList?CusNO=3481&NO=743	Electronic Resource	Other	Other
1087	Huang, C. (2005). Designing high-quality interactive multimedia learning modules. Computerized Medical Imaging and Graphics, Volume 29, Issue 2-3, Pages 223-233.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1102	Johnson, E. (2003). Meeting the challenge for sustainability. Bio-Link Connections newsletter. Volume 5, Issue 1.	Publication	STEM education/ research	Project Capacity Building
1114	Krasny, ME. (2005). University K-12 science outreach programs: How can we reach a broad audience? Bioscience, 55(4), 350-359.	Publication	STEM education/ research	Partnerships/systemic reform
1132	Lemtrakul, W. (2003). A Development of the Evaluation Method to Assess the Success of a Teacher Training Project Based on Theory – Based Evaluation.	Presentation	Evaluation	Evaluation
1137	Lövgren, B. (2005, October 20-21). Platsbesök as method wide evaluation of education. Paper to the Swedish Evaluation Association's conference. [Britta Lövgren, Platsbesök som metod vid utvärdering av utbildning. Paper till Svenska utvärderingsföreningens konferens 20-21 oktober 2005.]	Presentation	Evaluation	Evaluation
1169	Neil Evans, Peter Saflund, Dr. Manjari Wijenaike, John Engman, Kris Madura, & Fran Linhart. [National Workforce Center for Emerging Technologies and CompTIA] (2002). White Paper: IT Security Specialist:	Presentation	STEM education/ research	Workforce Development

	Integrating Academic Credentials with IT Professional Certifications. Paper presented at the Cyber-Security Education Workshop sponsored by the National Science Foundation and the American Association of Community Colleges.			
1171	Neves, P. (2004). NJCATE's National Engineering Technology Education Clearinghouse Facilitates Dissemination and Promotes Sustainability, Key Elements of ATE Projects. National Engineering Technology Education Clearinghouse (NETEC) Network online newsletter. http://www.neteconline.org/news/dec04-sharefindgreatideas.cfm	Electronic Resource	STEM education/ research	Project Capacity Building
1174	Patton, M., (Ed.) (2006). ATE Centers Impact 2006-2007.Tempe, AZ: Maricopa Community Colleges.	Report	STEM education/ research	Workforce Development
1181	Raine, E. (Unpublished thesis). Where Supply Meets Demand: Women in Diesel Mechanics. University of Kentucky.	Publication	STEM education/ research	Workforce Development
1182	Rebolloso, E., Fernández-Ramírez, B., Cantón, P. & Salvador, C. (2005). "Criterios de calidad de las evaluaciones universitarias" [Criteria of quality of the university evaluations] IV Conferencia de la Sociedad Española de Evaluación.	Presentation	Evaluation	Evaluation
1211	Sparling, P.B. (2005) Are wellness/fitness programs benefiting participants' movement and mobility in daily life? Quest, 57(1), 162-170.	Publication	Other	Other
1232	Wang Ying (2006). Senior U.S. technology education reform plan and the enlightenment to China. Vocational and Technical Education Forum, 23, 53-55.	Publication	STEM education/ research	Partnerships/systemic reform
1233	Washington State University Wiki "Program Evaluation Resources" http://wiki.wsu.edu/wsuwiki/Program_Evaluation_Resources	Electronic Resource	Evaluation	Evaluation
1244	Wesley, J. (2005). Current characteristics of faculty development in public two-year colleges in Texas. Doctoral dissertation. Texas A&M University.	Publication	Education - general	Professional Development
1249	Wood, N.B., Lawrenz, F., Huffman, D., Schultz, M. (2006). Viewing the school environment through multiple lenses:	Publication	STEM education/ research	Partnerships/systemic reform

	In search of school-level variables tied to student achievement Source: Journal of Research in Science Teaching, 43(3), 237 -254.			
1250	Xia Jianguo & Liu Xiao Bao. (2007). Application-typed Undergraduate Education: Backdrop and Essential. [Approximate translation from Chinese.] Research in Higher Education of Engineering, 3.	Publication	STEM education/ research	Workforce Development
1252	Yang, H.L., Shen, J.P., Cao, H.G., et al. (2004). Multilevel evaluation alignment: An explication of a four-step model. American Journal of Evaluation, 25(4), 493-507.	Publication	Evaluation	Evaluation
1256	Berg, D.H., Taylor, J., Hutchinson, N.L., Munby, H., Versnel, J. & Chin, P. (2007). Student assessment in exemplary work-based education programs. Journal of Workplace Learning, 19(4), 209-221.	Publication	Education - general	Workforce Development
1259	Bickman, L & Gerrard, W. (Spring 2004). Psy315b "Program Evaluation": Course Syllabus. Vanderbilt University.	Other	Evaluation	Evaluation
1260	Bio-link. (nd). Biotechnology Education. Available from: http://www.bio-link.org/skillstandards2.htm.	Electronic Resource	STEM education/ research	Workforce Development
1266	Bureau of Justice Assistance, Center for Program Evaluation "Useful Links". http://www.ojp.usdoj.gov/BJA/evaluation/links/index.htm	Electronic Resource	Evaluation	Evaluation
1272	Castellano, M., Stone, J.R., & Stringfield, S. (2005). Earning industry-recognized credentials in high school: Exploring research and policy issues. Journal of Career and Technical Education, 21(2).	Publication	STEM education/ research	Workforce Development
ATE001	Gullickson, A. R., Coryn, C. L. S., & Ritchie, L. A. (2006). Program evaluation. [aka "Project level evaluation practices"] [Advanced Technological Education Program Evaluation Briefing Paper Series, Briefing Paper #1]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Evaluation
ATE002	Coryn, C. L. S., Gullickson, A. R., & Ritchie, L. A. (2006). Contributors and inhibitors influencing program improvement [Advanced	Report	STEM evaluation	Project Capacity Building

	Technological Education Program Evaluation Briefing Paper Series, Briefing Paper #2]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.			
ATE003	Ritchie, L. A., Gullickson, A. R., & Coryn, C. L. S. (2006). Workforce needs assessment [Advanced Technological Education Program Evaluation Briefing Paper Series, Briefing Paper #3]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Evaluation
ATE004	Coryn, C. L. S., Gullickson, A. R., & Ritchie, L. A. (2006). Contributors and inhibitors influencing articulation agreements [Advanced Technological Education Program Evaluation Briefing Paper Series, Briefing Paper #4]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Project Capacity Building
ATE005	Ritchie, L. A., Gullickson, A. R., & Coryn, C. L. S. (2006). Challenges and resolutions [Advanced Technological Education Program Evaluation Briefing Paper Series, Briefing Paper #5]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development
ATE005	Ritchie, L. A., Gullickson, A. R., & Coryn, C. L. S. (2006). Challenges and resolutions [Advanced Technological Education Program Evaluation Briefing Paper Series, Briefing Paper #5]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development
ATE006	Gullickson, A. R., Coryn, C. L. S., & Ritchie, L. A. (2006). Materials development, professional development, and program improvement: Productivity and quality [Advanced Technological Education Program Evaluation Briefing Paper Series, Briefing Paper #6]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Materials Development
ATE018	Gullickson, A. R., & Tressler, G. R. (2002). Recruitment and retention. In A. R. Gullickson, F. P. Lawrenz, & N.	Publication	STEM evaluation	Workforce Development

	Keiser (Eds.), The ATE program: Issues for consideration: A monograph (pp. 103-129). Washington, DC: National Science Foundation.			
ATE019	Gold, N., & Powe, K. W. (2002). Assessing the impact and effectiveness of professional development. In A. R. Gullickson, F. P. Lawrenz, & N. Keiser (Eds.), The ATE program: Issues for consideration: A monograph (pp. 81-102). Washington, DC: National Science Foundation.	Publication	STEM evaluation	Professional Development
ATE025	Zinser, R. & nHannsen, C. (2006). Improving Access to the Baccalaureate: Articulation Agreements and the National Science Foundation's Advanced Technological Education Program. Community College Review, 34(1), 27-43.	Publication	STEM evaluation	Workforce Development
ATE028	Zinser, R., & Lawrenz, F. P. (2004). New roles to meet industry needs: A look at the Advanced Technological Education program. Journal of Vocational Education Research, 29(2), 85-100.	Publication	STEM evaluation	Workforce Development
ATE029	Lawrenz, F. P., Keiser, N., & LaVoie, B. (2003). Sustaining innovation in technological education. Community College Review, 30(4), 1-14.	Publication	STEM education/ research	Project Capacity Building
ATE030	Lawrenz, F. P., Keiser, N,.& LaVoie, B. (2003) Evaluative site visits: A methodological review. American Journal of Evaluation, 24(3), 341-352.	Publication	Evaluation	Evaluation
ATE032	Gullickson, A. R., Wingate, L., Lawrenz, F. A., & Coryn, C. L. S. (2006). The National Science Foundation's Advanced Technology Education Program: Final evaluation report. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development
ATE037	Appleton, J. J., & Lawrenz, F. P. (2005). ATE materials outcome comparison: Technical report. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Materials Development
ATE038	Appleton, J. J., & Lawrenz, F. P. (2004). ATE materials development process report. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Materials Development
ATE039	Keiser, J., Lawrenz, F. P., & Appleton,	Report	STEM evaluation	Materials

	J. J. (2004). Evaluation of materials produced by the ATE program. Kalamazoo, MI: Western Michigan University, The Evaluation Center.			Development
ATE062	Migotsky, C., & Stake, R. (2001). An evaluation of an evaluation: CIRCE's metaevaluation of the ATE program evaluation. Urbana-Champaign, IL: University of Illinois at Urbana-Champaign.	Report	STEM evaluation	Evaluation
ATE087	Hanssen, C. E., & Coryn, C. L. S. (2004, November). Theory-driven evaluations of multi-site programs. Paper presented at the meeting of the American Evaluation Association, Atlanta, GA.	Presentation	Evaluation	Evaluation

Endnotes

[i] Project "leaders" refers to project primary investigators, co-primary investigators, coordinators, directors, administrators, or project consultants.

[ii] Number of respondents who provided a response directly related to the question or program evaluation. Some responses referred to the survey itself or indicated that they did not have a comment by writing "none," or "no comment;" these types of responses were not considered direct and are not reported in this table.

[iii] ATE Final Report, 2006, p. vi

[iv] Gullickson, Lawrenz, & Keiser, 2000.

[v] ATE Final Report, 2006, p. 33.

[vi] The Sustainability of Advanced Technological Education Supported Efforts: An Evaluation Wayne W. Welch, University of Minnesota (ret.) and Arlen R. Gullickson, Western Michigan University August 2006, p.2, accessed htttp://www.wmich.edu/evalctr/ate/ATESustainabilityReport.pdf

[vii] ATE Program Evaluation Project. Presentation accessed WMU website at http://www.wmich.edu/evalctr/ate/presentations/gettingstartedfinal.pdf

[viii] NSF, 03 Program Solicitation, p. 2

[ix] ATE Program Evaluation Project. Presentation accessed WMU website at http://www.wmich.edu/evalctr/ate/presentations/gettingstartedfinal.pdf

[x] ATE Final Report, 2006.

[xi] ATE Final Report, 2006, p. 16.

[xii] http://www.wmich.edu/evalctr/ate/question.html

[xiii] ATE Program Evaluation Project. Presentation accessed WMU website at http://www.wmich.edu/evalctr/ate/presentations/gettingstartedfinal.pdf

[xiv] ATE Final Report, 2006, p. 14.

[xv] ATE Final Report, 2006, p. 15.

[xvi] A Guide for Planning and Implementing Site Visits, Frances Lawrenz, University of Minnesota, Nanette Keiser, Western Michigan University, and Beth Lavoie, University of Minnesota. March 2002.

[xvii] The ATE Program: Issues for Consideration by Frances Lawrenz and Nanette Keiser, Winter 2002, p.9. http://www.wmich.edu/evalctr/ate/ATEpapers/issuesforconsider.pdf

[xviii] ATE Program Evaluation Products. The Evaluation Center, Western Michigan University. Accessed 2/15/07 at http://www.wmich.edu/evalctr/ate/evalproducts.htm.

[xix] Welch, W & Robert Reineke. National Visiting Committee (NVC) Handbook. 2003. The Evaluation Center, Western. Accessed 2/15/07 at http://www.wmich.edu/evalctr/ate/evalproducts.htm.

[xx] Gullickson, Arlen. PI Reflections, personal communication

[xxi] All means were calculated by assigning scores as follows: 1=No involvement, 2=Yes, a little, 3=Yes, some, 4=Yes, extensively.